



The ICEE Conference 2025

July 8-11th, 2025 Wuhan, China

A Survey on Inertia Estimation Algorithm for Frequency Control and a Kalman Filter Approach

Name: Karl M.H. LAI

Company/University: CLP Power Hong Kong / HKU

Country: Hong Kong, China





A Survey on Inertia Estimation Algorithm for Frequency Control and a Kalman Filter Approach

Karl M.H. LAI

Content

- Frequency Events in Power Systems – Introduction
 - Different Frequency Response, Different Activation – How to Coordinate?
 - How much Energy on Hand?
 - How to Govern Frequency Control?
 - What Frequency to Restore?
 - Inertia Estimation – Challenge
- Inertia Estimation Algorithm (IEA)
 - Disturbance-based Approach
 - Single Machine vs Multi-machine Approach, Centre of Inertia (COI)
 - How to Filter Frequency?
 - How to Eliminate Droop and Load Damping Effect?
 - Energy-Conservation Approach
 - Least Square Solution
 - Electromechanical Mode
 - System Identification Approach – ARMAX
 - Recursive Least Square (RLS) Approach
- Extended Kalman Filter (EKF) Approach – Analytical Linearization
- Practical Solution

Frequency Events in Power System – Introduction

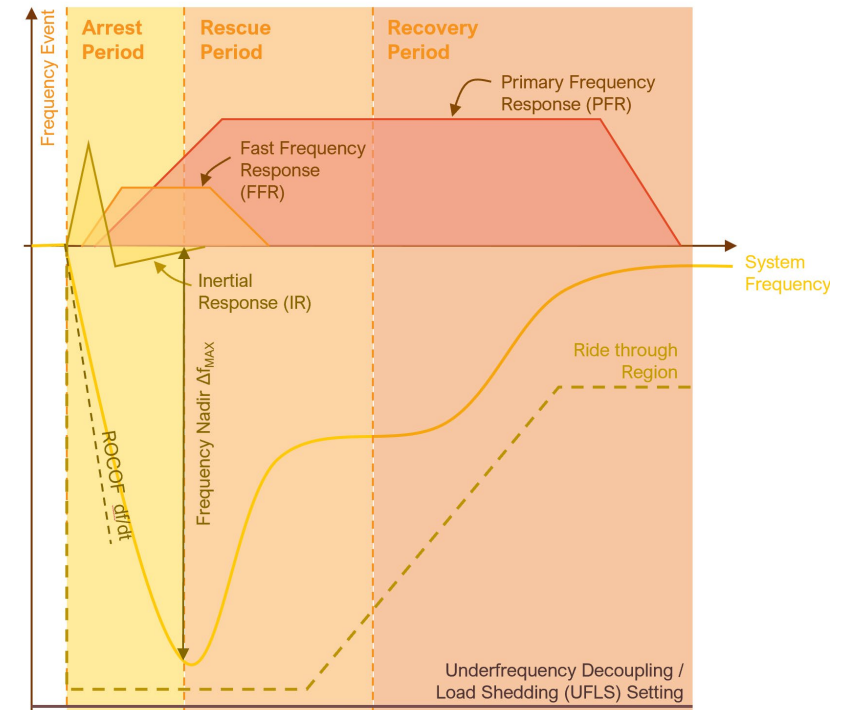
- **Power System Frequency** ($f_s = f_n + \Delta f$ [Hz]) is dependent on the **net power exchange** ($\Delta P = P_M - P_L$) between generation and load. Frequency events are the events which can cause the power system to reach **abnormal frequency**, in which may cause equipment damage such as for the machines.
- **Possible Frequency Events** are –
 - Load Rejection
 - Load Step (Upward or Downward)
 - Generation Change (due to intermittent nature of RE)
 - Faults (reduced voltage leading to incapable of power transfer)

- **Why Frequency Events is now an issue?**

- ↑ Non-synchronous generation
- ↑ HVDC infeed

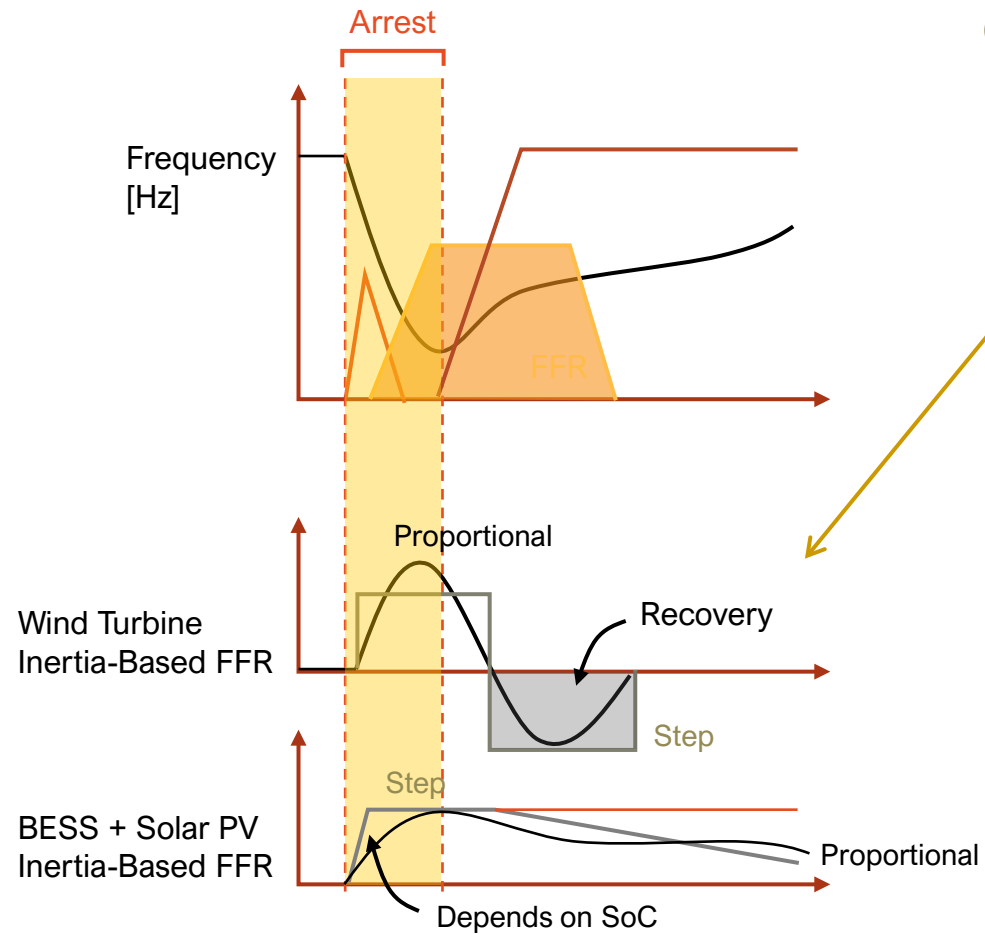
Hence,

- ↓ System inertia (H)
- ↑ Frequency Nadir and ROCOF
- Leading to
 - **Cascade tripping**
(due to protection setting, large frequency deviation, incorrect grid partition)
 - **Failure of UFLS scheme**
(due to measurement delay in triggering UFLS)
 - **Equipment failure**
(e.g. generator shaft due to shock load with large ROCOF)



General Issues in *Modern* Frequency Control

1. Different Frequency Response Component, Different Activation → How to Coordinate?

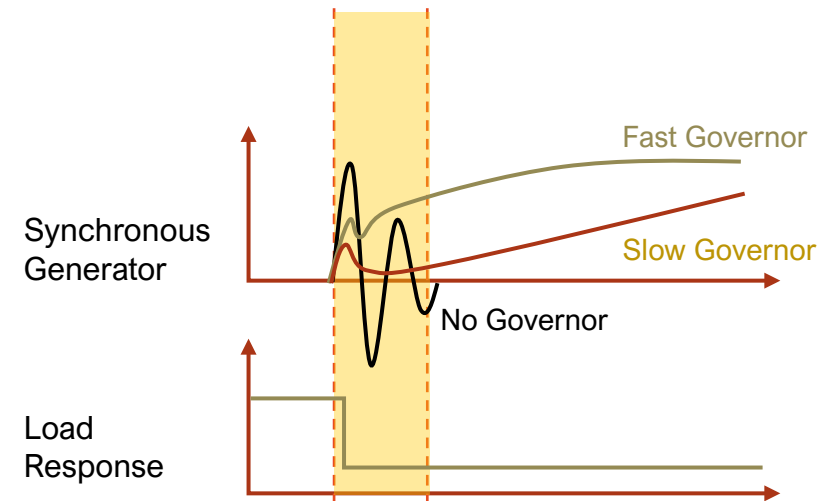


Check 1) Response Speed
2) Response Shape
3) Response Duration



a) Inertia (recovery / compensation)
b) PFR (delta control needed)
c) FFR (pre-charge needed)

**NOT Optimal Power Flow (OPF) Issue only,
but an Optimal Control Issue in Real Time**



General Issues in *Modern* Frequency Control

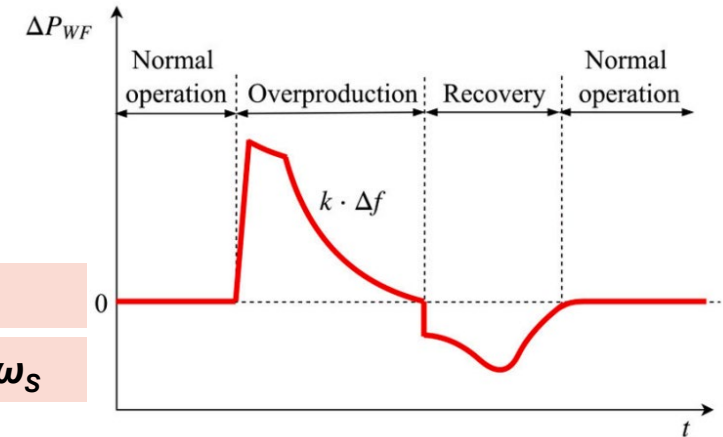
2. How much energy on hand?

- PV → Through Pre-curtailment → In what cost?
- BESS → Through Pre-Charging and SOC Estimation → In what cost?

- Wind → Through 1) Pre-curtailment by overspeeding
2) Extra Extraction with Pitching
3) Releasing KE from Turbine
4) Additional Source from BESS

→ **Limit Energy Use** - $|E - P\Delta t|$

→ **Limit Both Frequency** Ω_m and ω_s



- **Need local information from grid to determine how to respond.**

→ **Inertia Estimation Problem.**

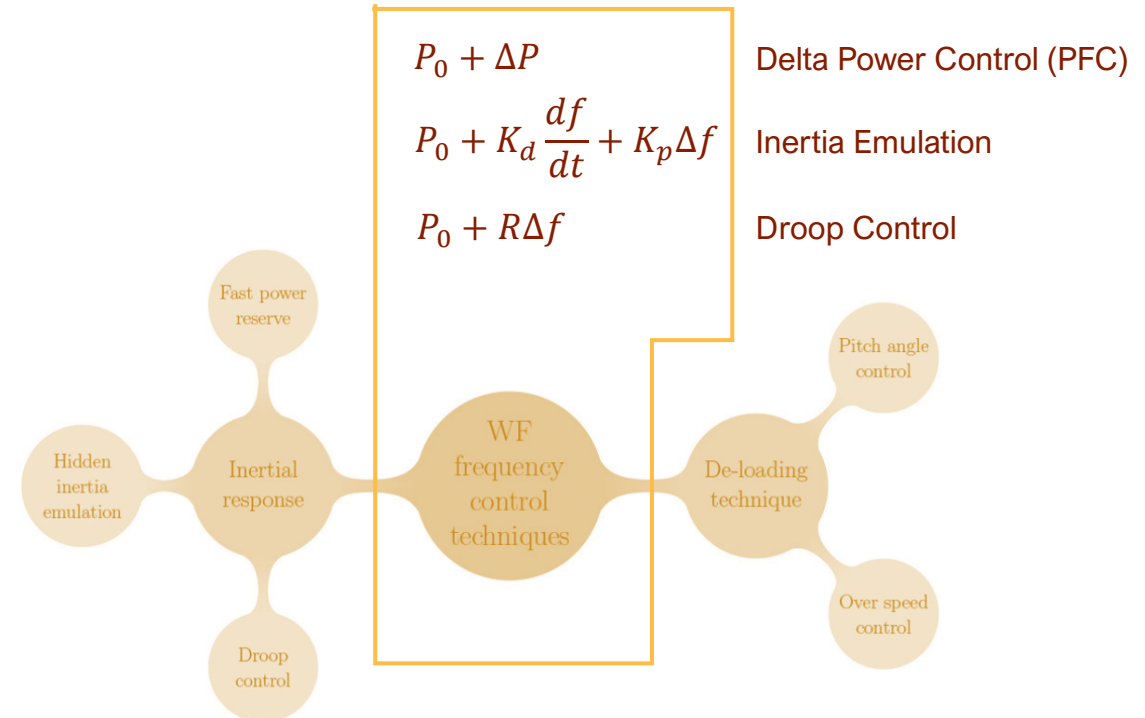
$$\frac{d\Delta f}{dt} = \frac{1}{2H_{eq}} (\Delta P_M - \Delta P_L - D\Delta f) \quad H_{V,eq} = 0.0357 \text{ (Wind \%)}$$

$$H_{eq} = \frac{\left(\frac{H_{R,eq}}{\sum_{i=1}^{CP} H_i S_{B,i}} + \frac{H_{V,eq}}{\sum_{j=1}^{VG} H_{V,j} S_{B,j}} \right)}{S_B} \quad \text{Wind Turbine Component, Time Dependent}$$

Inertia Estimation

$$H_{eq} = - \frac{\Delta P}{\frac{2d\left(\frac{\Delta f}{f_0}\right)}{dt} \Big|_{t=0^+}} = \frac{R(t_s)\Delta f(t_s)}{2\frac{df(t_s)}{dt}} - \frac{\Delta P_{LV}(t_s)}{2\frac{df(t_s)}{dt}} - \Delta P_{dist}$$

Governor Frequency Response
Load Voltage Response
Disturbance



General Issues in *Modern* Frequency Control

Require **adequate energy buffer** and **control system** in place

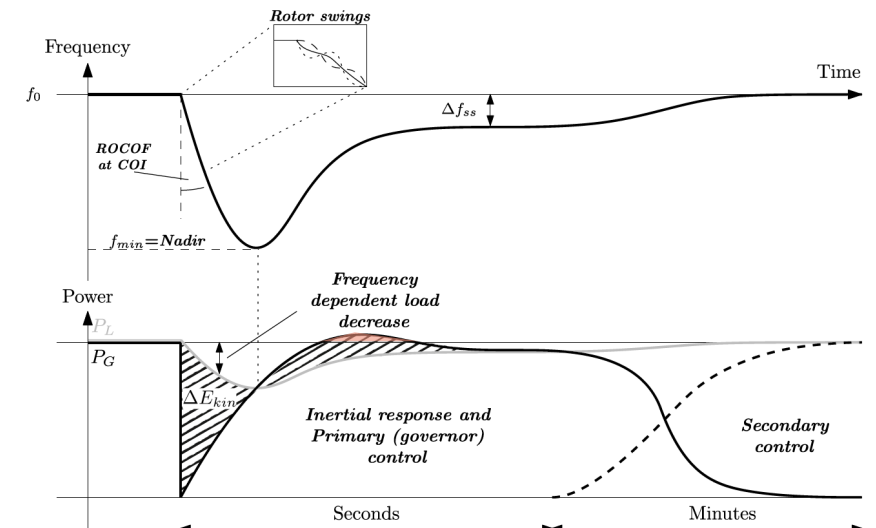
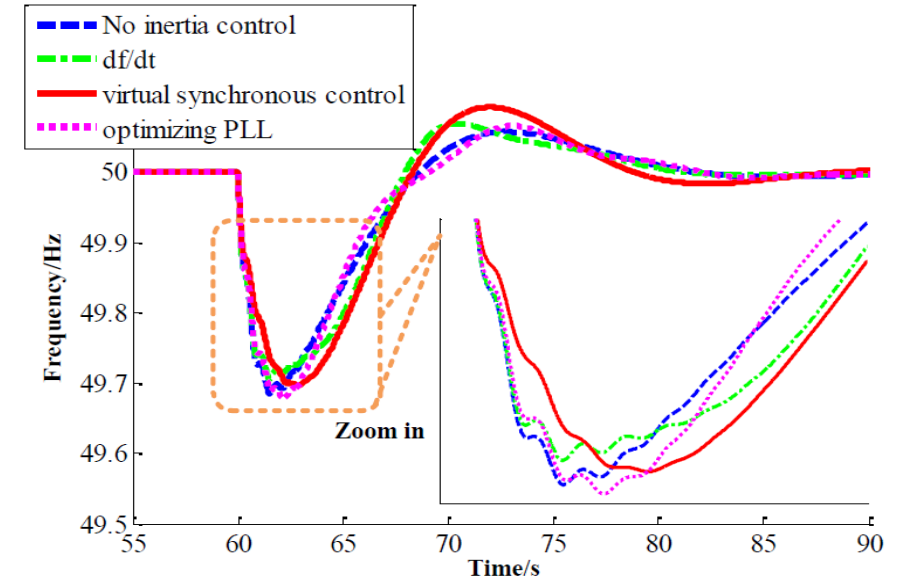
- To arrest and rescue frequency deviation
- **Limited inertia emulation** due to
 - **No “appropriate” regulation** of inertia emulation / percentage of inertia from IBR
 - **Unknown impact of system dynamics** due to control characteristics of inertia emulation
 - **Uncertain energy buffer** leads to different response
- **Goal – to minimize ROCOF and UFLS tripping –**

→ **Limit Energy Use - $|E - P\Delta t|$**

$$\frac{2H}{f_0} \frac{d\Delta f}{dt} + D\Delta f = \Delta P^G + \Delta P^W - \Delta P_e$$

→ **Limit Both Frequency Ω_m and ω_s**

- **Change grid code requirement** (connection criteria) for IBR plants
- **Modify protection relay setting** to cope with higher ROCOF and larger frequency swing
 - # check **frequency reserve contract in real time**
- **Reduce power imbalance ΔP_e** , i.e. “single largest contingency”
 - (often the largest generator connected)
 - # check **ramp rate limit, time for synchronization, network constraints**
- **Increase inertia response H** by having more balancing mechanism
 - # check **start-up cost, hot-standby payment, virtual inertia trading**



General Issues in *Modern* Frequency Control

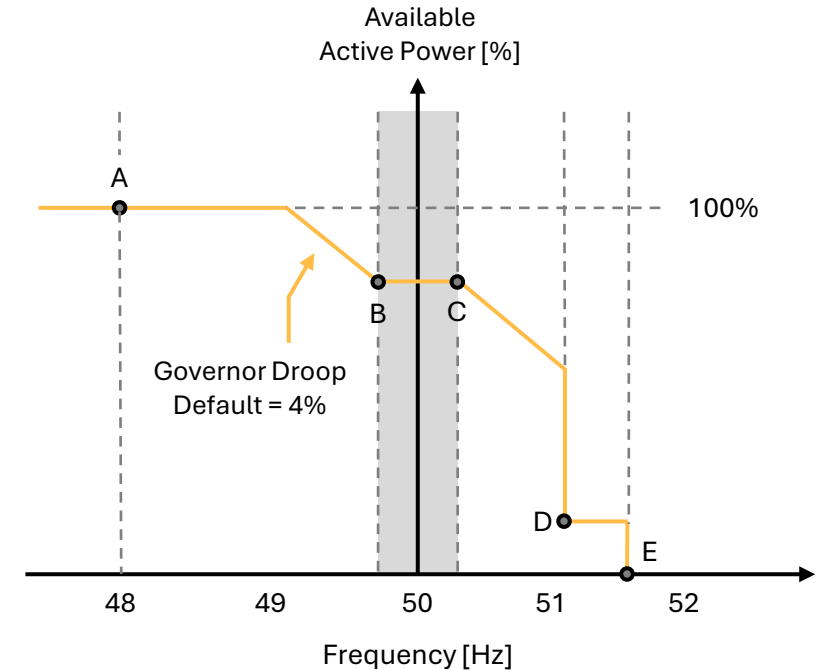
3. How to Govern Frequency Response?

Grid code requirements are often formulated in terms of

- droop percentage [or Hz/MW] with an associated deadband
- inertia requirement [s], or
- activation requirement

The following are examples of the grid codes. It should be noted that certain utilities, such as those in Hong Kong, may not impose compulsory requirements on frequency support.

- **UK:** Deadband = 40.5 – 50.5Hz; PFR droop = 3% - 5%
- **Ireland:** PFR droop = 2% - 10% as controlled by TSO (default = 4%)
- **Germany:** PFR = > 2%; Only required with WPPs ≥ 100 MW
- **Quebec:** Inertia Constant = 3.5s; Only required with output > 10MW



General Issues in *Modern* Frequency Control

4. What Frequency to be Restored?

Frequency Estimation Algorithms

1. Zero Crossing

Vulnerable to disturbance (e.g. DC offset) and harmonics

2. Discrete Fourier Transform (DFT)

Assumed Stationary Signal

3. Adaptive Notch Filter (ANF) and Phase-Locked Loop (PLL)

Robust to noise and easy to implement

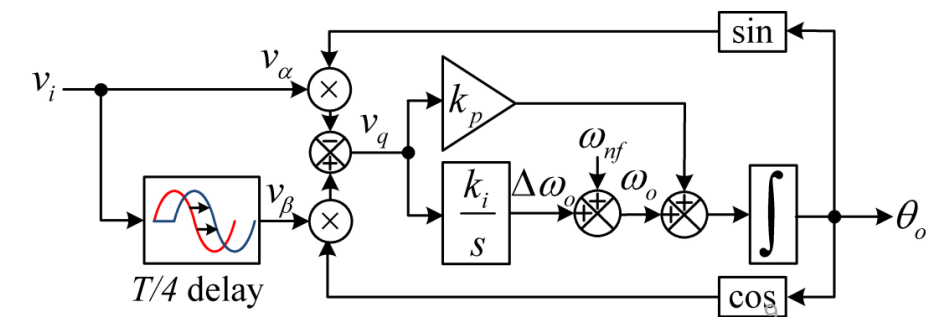
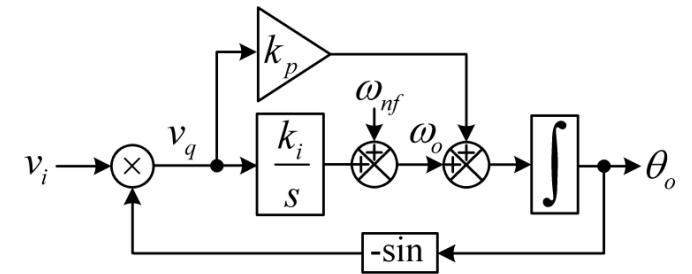
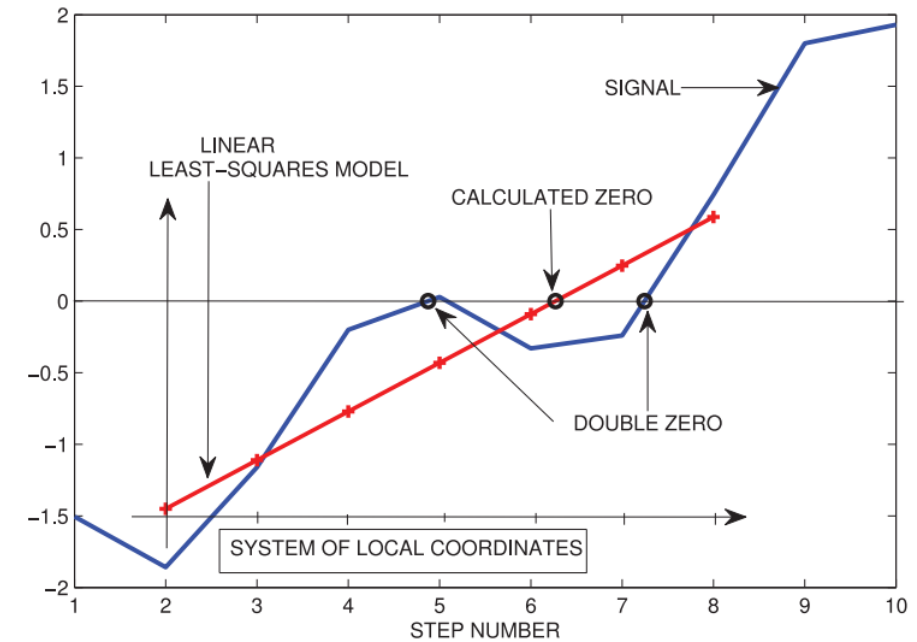
4. Others

Types of PLL / FLL: e.g. Quadrature PLL, SOGI – PLL

Types of Filter – e.g. Volterra Integral + State Observer

Challenge in *Meaningful* Frequency Estimation –

- Frequency Variation
- Noise
- Harmonics
- Additional Zero Crossing
- Amplitude Step
- Phase Jump



Inertia Estimation - Challenge

Dynamics for Independent SG:

$$2H_{SG,i} \frac{d\Delta\omega_{g,i}}{dt} = \Delta P_{m,i} - \Delta P_{e,i} - D_i \Delta\omega_{g,i}$$

$$H_{eq} = \frac{\sum_{i=1}^{N_{SG}} \overbrace{H_{SG,i} S_{B,i}}^{(SGs)} + \sum_{j=1}^{N_{CIG}} \overbrace{H_{CIG,j} S_{B,j}}^{(CIGs)}}{\sum_{i=1}^{N_{SG}} S_{B,i} + \sum_{j=1}^{N_{CIG}} S_{B,j}}$$

Equivalent Model:

$$2H_{eq} \frac{d\Delta\omega_{eq}}{dt} = \Delta P_m - \Delta P_e - D_{eq} \Delta\omega_{eq}$$

Aggregated Solution -

The inertia Constant of different types of power plants.

Power plant type	Inertia Constant (s)
Hydro	1.75–4.75
Nuclear	4
Thermal	2–10

Control Dependent -

Synthetic Inertia:
(Droop Control)

$$P_c = P_{ref} + \Delta P = P_{ref} - K_d \frac{df_b}{dt} - K_e (f_b - f_n)$$

Virtual Inertia:

$$P_{in} - P_{out} = 2H_{VSG} \frac{d\omega_{VSG}}{dt} + D_{VSG} \Delta\omega_{VSG}$$

Known Inertia from all source
= Know equivalent Inertia

Frequency / Voltage Dependent Load + Disturbance -

$$2H_{eq} \frac{d\Delta\omega_{eq}}{dt} = - (\Delta P_L + \Delta P_{dist})$$

Problem:

1. What is f_{sys} ?
2. What is ΔP_E ?
3. What is H_{eq} ?

Average Frequency:

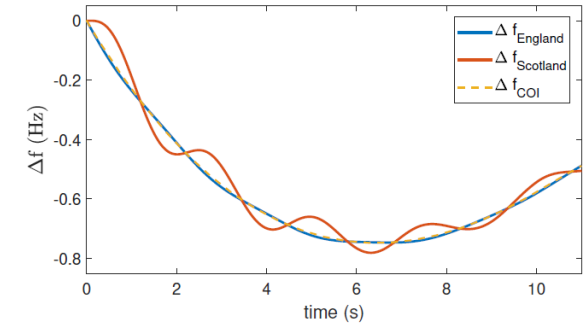
$$\tilde{f}(t) = k[1 + \alpha e^{-\beta\omega_1 t} \sin(\omega_2 t + \phi)]$$

Load Step due to Voltage:

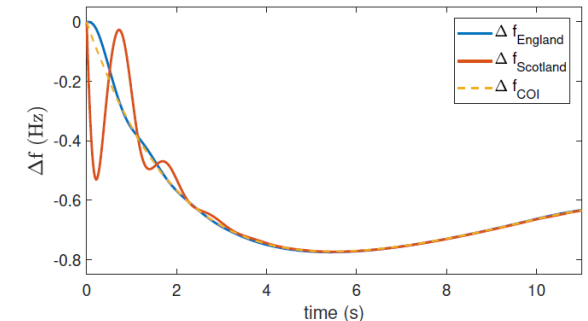
$$\Delta P_L = P_{L0}(V_s - 1)$$

Definition: $H_i = \frac{E_{ki}}{S_{r_i}} = \frac{\frac{1}{2} J_i \omega_{m0,i}^2}{S_{r_i}}$

Frequency of Separated Grids -



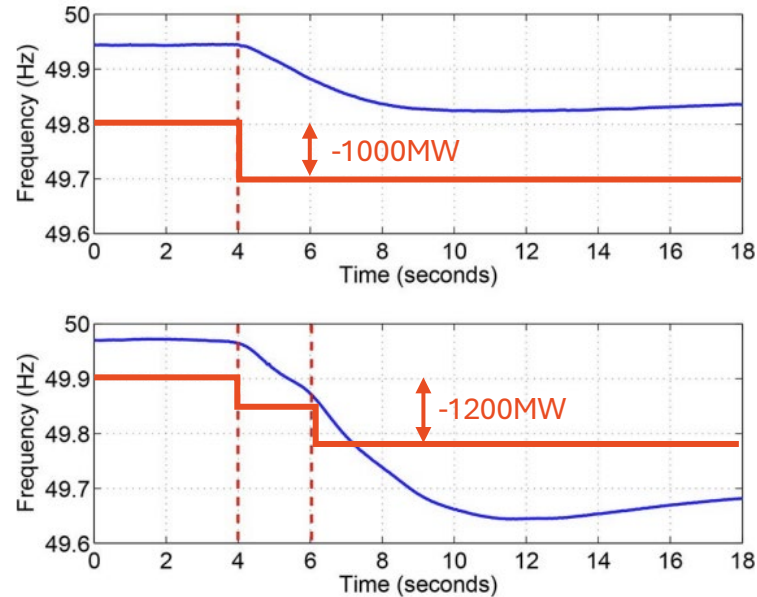
(a) 1.8GW generation loss in England



(b) 0.6GW generation loss in Scotland

$$f_{Col} = \frac{\sum_{i=1}^{N_{SG}} H_i f_i}{\sum_{i=1}^{N_{SG}} H_i}$$

Inertia Estimation - Disturbance-based Approach



Factor Affecting Inertia Estimation -

- precise data on the size of loss;
- online plant inertias (for estimate of residual contribution);
- identification of **event start time**;
- accuracy of frequency measurement;
- method of RoCoF calculation;
- location of measurement point relative to in-feed loss

Swing Equation -

$$2H_{total} \frac{d\Delta f}{dt} = -\Delta P_d - \Delta P_l$$

Require Detrended Fluctuation Analysis (DFA) by T. Inoue (1997) or Second Order Method by W. Wang (2020) to identify the start time (T0)

Toshio Inoue (1997): Polynomial Filtering -

$$\frac{\Delta f}{f_r} = A_5 t^5 + A_4 t^4 + A_3 t^3 + A_2 t^2 + \boxed{A_1} t$$

$$H_{Total} = \frac{1}{2S_r} \frac{-\Delta P_e}{A_1}$$

D. Zografos (2018): RV Approach -

$$H_{est} = \frac{h_1(f(t_s)) + h_2(V(t_s)) - \Delta P_{dist}}{2(df(t_s)/dt)}$$

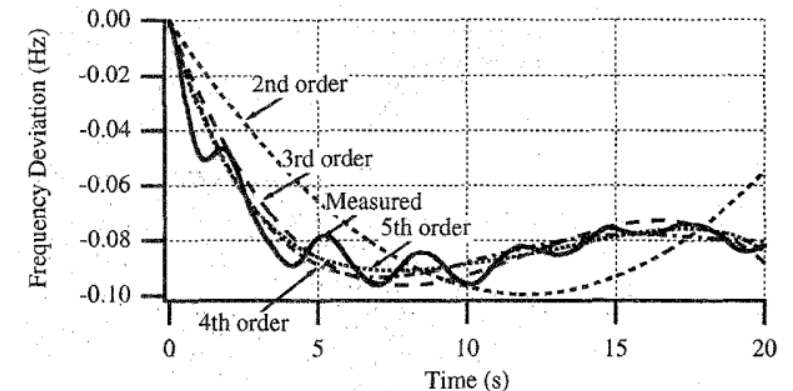
$$= \frac{R(t_s)\Delta f(t_s) - \Delta P_{LV}(t_s) - \Delta P_{dist}}{2(df(t_s)/dt)}$$

D. del Giudice (2019): Window Based Approach -

$$H = \frac{1}{2} \frac{P_e(t_0^-) - P_e(t_0^+)}{\frac{\partial f(t_0^+)}{\partial t} - \frac{\partial f(t_0^-)}{\partial t}} = \frac{1}{2} \frac{p_1 - p_2}{r_2 - r_1}$$

$$p_1 = \frac{\sum_{i=t_0-At_s}^{t_0} P_e(t)}{A} \quad r_1 = \frac{\sum_{i=t_0-At_s}^{t_0} \frac{\partial f(t)}{\partial t}}{A}$$

$$p_2 = \frac{\sum_{i=t_0+Wt_s}^{t_0+W\Delta t+At} P_e(t)}{A} \quad r_2 = \frac{\sum_{i=t_0+Wt_s}^{t_0+W\Delta t+At} \frac{\partial f(t)}{\partial t}}{A}$$



Inertia Estimation – Disturbance Based Approach

Anderson & Fouad (2008) – Swing Equation Based (Single Machine)

Determine inertia of each generators and sum up -

$$H_i = \frac{\Delta P_i \cdot f_n}{2 \cdot \left(\frac{df}{dt}\right)_i \cdot S_{rated}} \rightarrow H_{system} = \sum_{i=1}^n H_i$$

Sharma (2011) – Swing Equation Based (Aggregated Model)

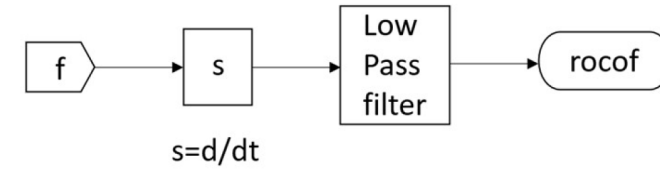
$$H_{system} = \frac{\sum_{i=1}^n (\Delta P_{mi} - \Delta P_{ei}) \cdot f_n}{2 \cdot \frac{df_{COI}}{dt} \cdot S_{rated}}$$

D. Zografos (2018) – RV Approach -

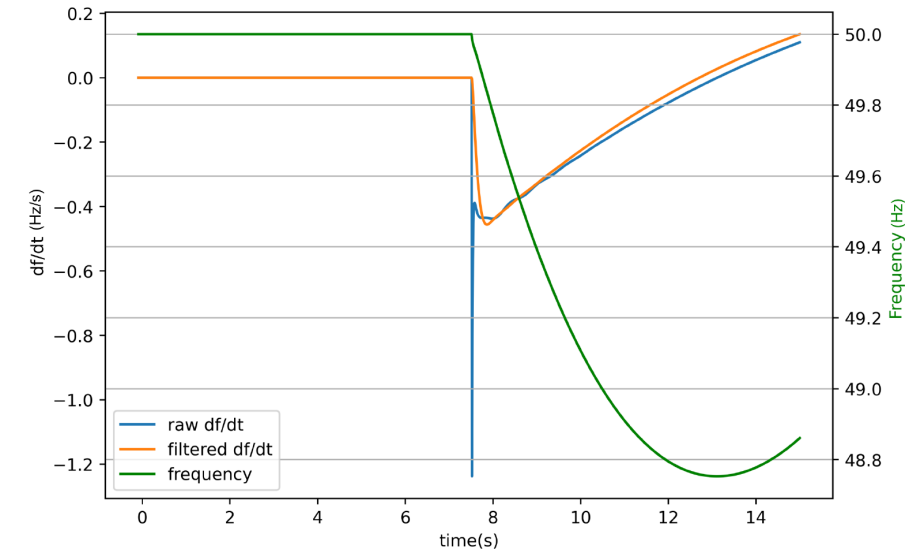
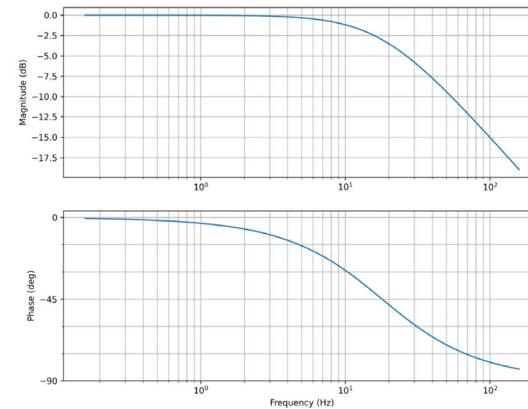
$$H_{system} = \frac{R(t)\Delta f(t) - \Delta P_{LV}(t_s) - \Delta P_{dist}}{2 \cdot \frac{df_{COI}}{dt}}$$

$$P_{LV}(t) = \sum_{i=1}^l P_{LO_i} \left(k_Z \left(\frac{V_{L_i}(t)}{V_{LO_i}} \right)^2 + k_I \left(\frac{V_{L_i}(t)}{V_{LO_i}} \right) \right)$$

ROCOF for Inertia Estimation



$$T(s) = \frac{1}{1 + \left(\frac{s}{\omega_o}\right)}$$



Inertia Estimation – Disturbance Based Approach

Yang (2021) – Disturbance Based (to eliminate effect of droop and damping)

$$f_{coi} = \frac{\sum_{i=1}^n H_i f_i}{\sum_{i=1}^n H_i}$$

$$2H \times \frac{d\Delta f}{dt} = P_m - P_e - D\Delta f$$

$$2H_{\Sigma} \times \frac{d\Delta f_{coi}}{dt} = P_{\Sigma SG} + P_{\Sigma RE} - P_{\Sigma L} - P_{\Sigma Loss}$$

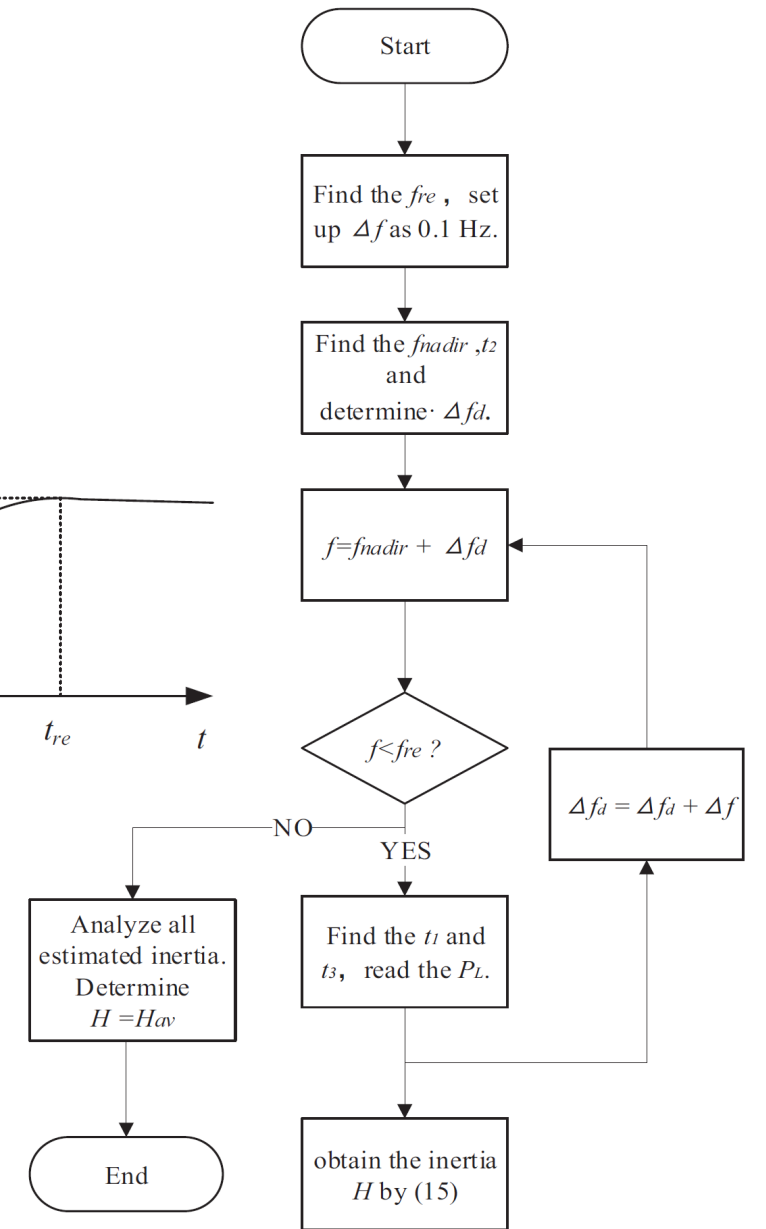
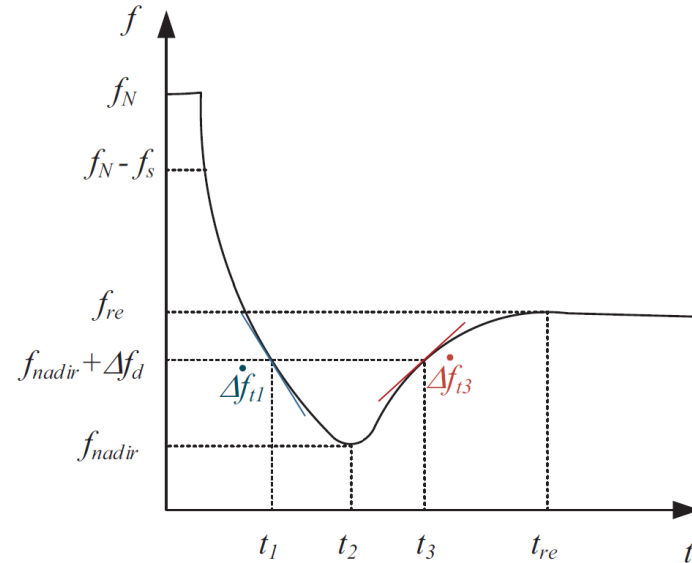
$$\begin{cases} \Delta P_{\Sigma SG} = P_{SG,1} + P_{SG,2} + \dots + P_{SG,n} \\ \quad = -(K_1 + K_2 + \dots + K_n) \times \Delta f_{coi} \\ \quad = -K_{SG} \Delta f_{coi} \\ \Delta P_{\Sigma RE} = -K_{RE} \Delta f_{coi} \end{cases}$$

$$f_{t_1} = f_{t_3} = f_{t_2} + \Delta f_d = \text{Equal Frequency Difference}$$

$$\Delta f_d \in [0, f_{nadir} - f_{re}]$$

$$\begin{aligned} & 2H \times (\Delta \dot{f}_2 - \Delta \dot{f}_1) \\ &= (P_{\Sigma SG|t_2} - P_{\Sigma SG|t_1}) + (P_{\Sigma RE|t_2} - P_{\Sigma RE|t_1}) - (P_{\Sigma L|t_2} - P_{\Sigma L|t_1}) \\ -) & 2H \times (\Delta \dot{f}_3 - \Delta \dot{f}_2) \\ &= (P_{\Sigma SG|t_3} - P_{\Sigma SG|t_2}) + (P_{\Sigma RE|t_3} - P_{\Sigma RE|t_2}) - (P_{\Sigma L|t_3} - P_{\Sigma L|t_2}) \end{aligned}$$

$$\rightarrow H = \frac{1}{2} \frac{P_{\Sigma L|t_3} - P_{\Sigma L|t_1}}{\Delta \dot{f}_3 - \Delta \dot{f}_1}$$



Inertia Estimation – Energy Conservation Approach

Wang (2020) - Energy Conservation Approach

System Equivalent Rotor Motion: [Actual Base]

$$\frac{2HS_B}{f_0} \frac{d\Delta f_{COI}}{dt} = \Delta P_m(t) - \Delta P_e(t) = -\Delta P_H(t)$$

From occurrence of disturbance to frequency nadir:

$$-\frac{2E_k}{f_0} \int_{t_0}^{t_2} \frac{d\Delta f_{COI,2}}{dt} dt = \int_{t_0}^{t_2} \Delta P_H dt = \Delta E_H$$

Energy Released from Inertial Response Divided into Two Stages:

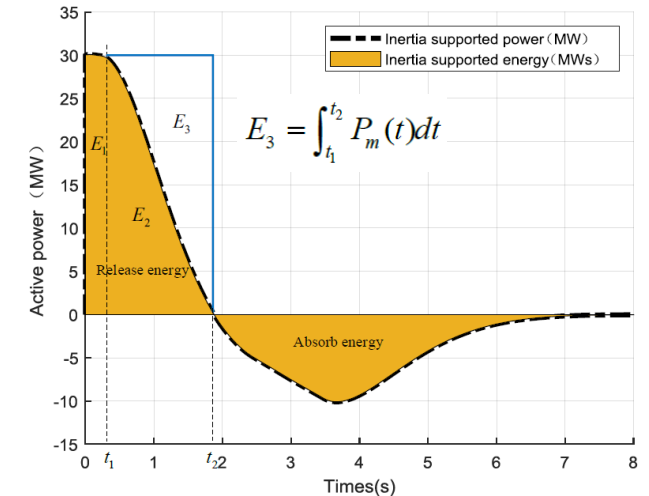
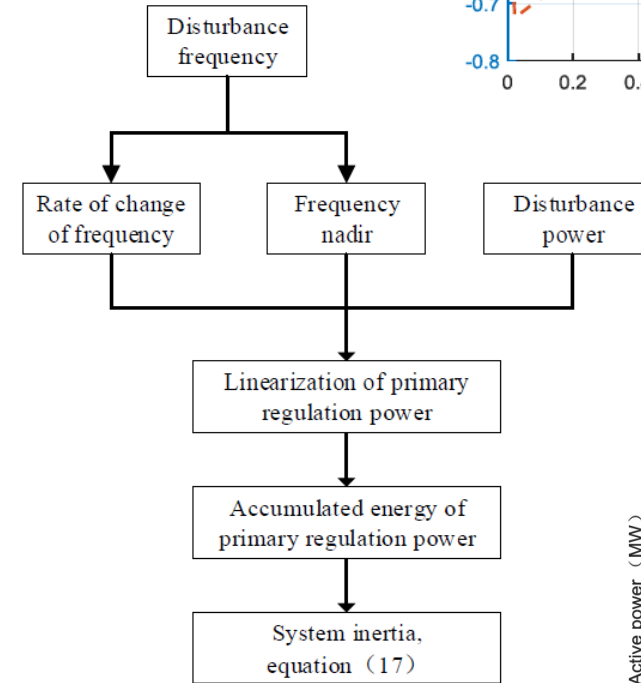
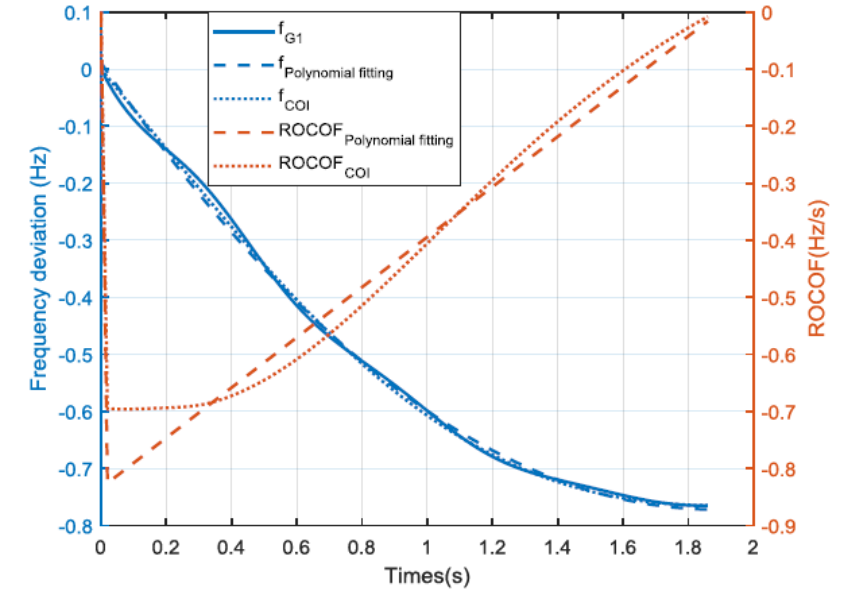
$$-\frac{2E_k}{f_0} \Delta f_{COI,2} = \int_{t_0}^{t_2} \Delta P_d dt - \int_{t_1}^{t_2} \Delta P_m(t) dt$$

System Inertia Expression: [Note: $E_k = HS_B$]

$$E_k = -\frac{f_0}{2\Delta f_{COI,2}} \left(\int_{t_0}^{t_2} \Delta P_d dt - \int_{t_1}^{t_2} \Delta P_m(t) dt \right)$$

Cumulative Energy
of Generator PFR
Cumulative Energy
of Disturbance
Power at stage $t_0 \sim t_2$

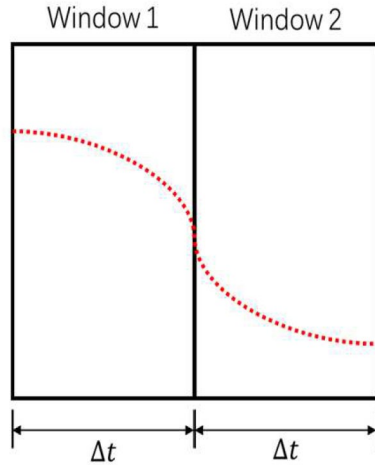
$$\Delta P_m(t) = \Delta P_d + \frac{2HS_B}{f_0} \frac{d\Delta f_{COI}}{dt} \quad \frac{d\Delta P_m(t)}{dt} = -\frac{d\Delta P_H(t)}{dt} = \frac{2HS_B}{f_0} \frac{d^2\Delta f_{COI}}{dt^2}$$



Inertia Estimation – Weight Least Square Method

Y. Wang (2023) – WLS Method

At k-th Sampled Point in m-th Sampling Window:



$$2H \frac{df_{m,k}}{dt} + D\Delta f_{m,k} = P_{M_{m,k}} - P_{E_{m,k}}$$

Difference of k-th Sample between first window and second window:

$$2H\Delta \frac{df}{dt}_{2-1,k} + D\Delta f_{2-1,k} = \Delta P_{M_{2-1,k}} - \Delta P_{E_{2-1,k}}$$

Minimize the 2-norm of the residual -

Minimize:

$$\sum_{k=1}^n \left\| 2H_{est}\Delta \frac{df}{dt}_{2-1,k} + D\Delta f_{2-1,k} - R(P_m)\Delta t + \Delta P_{E_{2-1,k}} \right\|$$

Consider 3-DoF Problem (including Change of Mechanical Power) -

$$\begin{bmatrix} \Delta \frac{df}{dt}_{2-1,1} & \Delta \frac{df}{dt}_{2-1,2} & \cdots & \Delta \frac{df}{dt}_{2-1,n} \\ \Delta f_{2-1,1} & \Delta f_{2-1,2} & \cdots & \Delta f_{2-1,n} \\ -\Delta t & -\Delta t & \cdots & -\Delta t \end{bmatrix} \begin{bmatrix} 2H_{est} \\ D \\ R(P_m) \end{bmatrix} = \begin{bmatrix} -\Delta P_{E_{2-1,1}} & -\Delta P_{E_{2-1,2}} & \cdots & -\Delta P_{E_{2-1,n}} \end{bmatrix}^T$$

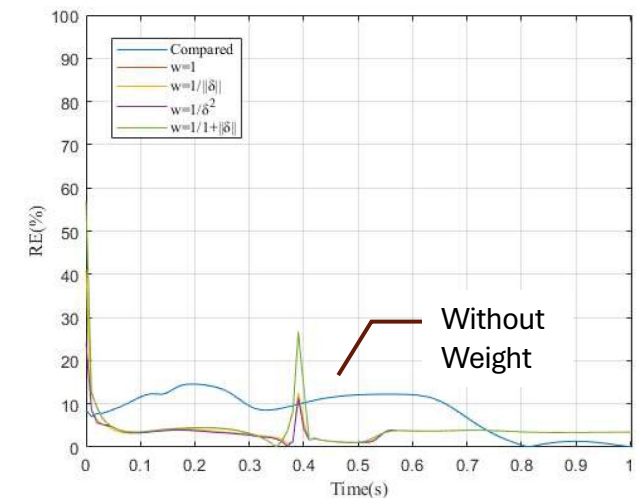
Least Square Solution: $X = \begin{bmatrix} 2H_{est} \\ D \\ R(P_m) \end{bmatrix} = (F^T F)^{-1} F^T P$

σ_i denotes the residuals of the i -th pair of sampling points. P_i and F_i are the i -th row of the corresponding matrix.

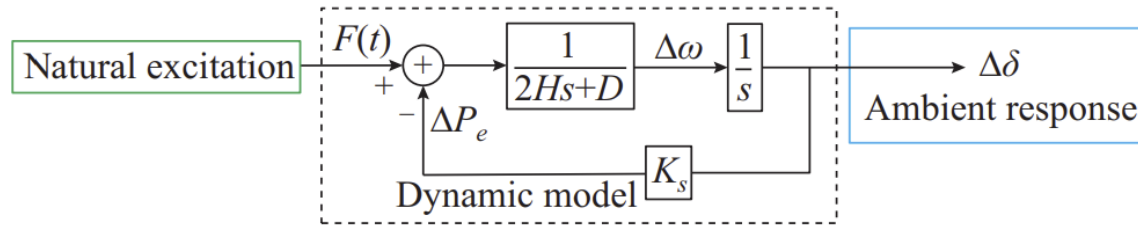
$$\sigma_i = P_i - F_i X$$

$$\begin{bmatrix} 2H_{est}' \\ D_{est}' \\ R(P_{M_{2-1}})' \Delta t \end{bmatrix} = (F^T W F)^{-1} F^T W P \quad \begin{matrix} P' = W^{\frac{1}{2}} P \\ F' = W^{\frac{1}{2}} F \end{matrix}$$

$$W = \begin{bmatrix} w_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & w_n \end{bmatrix} \quad w_i = \begin{cases} 1/\|\sigma_i\| \\ 1/\sigma_i^2 \\ 1/1 + \|\sigma_i\| \end{cases}$$



Inertia Estimation – Electromechanical Mode Estimation



Stochastic Subspace Identification (SSI):

Data Hankel
Matrix -

$$Y_{1|2i-1} = \begin{bmatrix} y_1 & y_2 & \cdots & y_j \\ y_2 & y_3 & \cdots & y_{j+1} \\ \vdots & \vdots & \ddots & \vdots \\ y_{2i-1} & y_{2i} & \cdots & y_{n_d-1} \end{bmatrix}$$

$$j = n_d - 2i$$

Unbiased Estimate -

$$\Lambda_{1|2i-1}^r = \frac{1}{j} Y_{1|2i-1} Y_{0|0}^{rT} = \begin{bmatrix} \Lambda_1^r \\ \Lambda_2^r \\ \vdots \\ \Lambda_{2i-1}^r \end{bmatrix}$$

Block Toeplitz
with SVD -

$$L_{1|i}^r = \begin{bmatrix} \Lambda_i^r & \Lambda_{i-1}^r & \cdots & \Lambda_1^r \\ \Lambda_{i+1}^r & \Lambda_i^r & \cdots & \Lambda_2^r \\ \vdots & \vdots & \ddots & \vdots \\ \Lambda_{2i-1}^r & \Lambda_{2i-2}^r & \cdots & \Lambda_i^r \end{bmatrix} = U_i \Sigma_i V_i^T = O_i C_i^r$$

Truncated
Data
Matrix -

$$A = \underline{O}_i^+ \underline{O}_i \quad f_k = \frac{|f_s \ln \lambda_k|}{2\pi} \quad \tilde{\zeta}_k = \frac{\text{Re}(\ln \lambda_k)}{|\ln \lambda_k|}$$

$$T_J \Delta \ddot{\delta}(t) + D \Delta \dot{\delta}(t) + K_s \Delta \delta(t) = F(t) \quad \text{Actual Dynamics}$$

Determine the
frequency and
damping from data

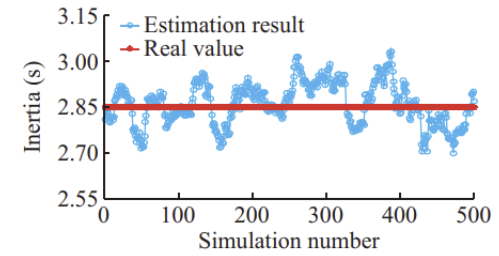
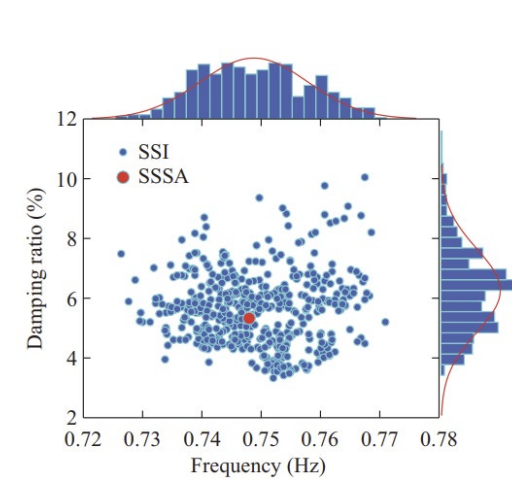
$$2\pi f_d = \frac{\sqrt{8\omega_0 H P_{e0} \cot \delta_0 - D^2}}{4H}$$

$$\zeta = \frac{D}{2\sqrt{P_{e0} \cot \delta_0 \cdot 2H\omega_0}}$$

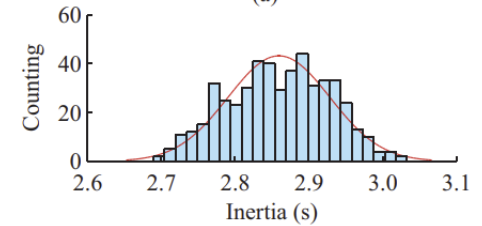
$$H = k_0 (1 - \zeta^2) f_d^{-2}$$

$$\Delta \ddot{\delta}(t) + 2\zeta \omega_n \Delta \dot{\delta}(t) + \omega_n^2 \Delta \delta(t) = F(t) \quad \text{2nd Order Dynamics}$$

Bo Wang (2022): SSI + EM Estimation



(a)



(b)

Method	Frequency (Hz)	Damping ratio (%)
SSSA	0.748	5.333
SSI	$\mu=0.759, Std=0.009$	$\mu=5.596, Std=1.171$

Model order	Reality	C-1	C-2	
			μ	Std
2 nd -order	2.85	2.855	2.859	0.0388
6 th -order	2.85	2.787	2.794	0.0382
6 th with exciter	2.85	2.916	2.921	0.0367

C-1: Calculate inertia of each sample and take the mean
C-2: Calculate the mean of all samples and determine inertia

Inertia Estimation – Electromechanical Mode with Dynamic Mode Decomposition (DMD)

Dynamic Model -

$$\Delta \dot{\theta} = \Delta \omega$$

$$M \Delta \dot{\omega} + D \Delta \omega = -P_e$$

Snapshot Matrix and Discrete State Evolution:

$$\mathbf{X}_{N-1} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \cdots \quad \mathbf{x}_{N-1}] \in R^{M \times N-1}$$

$$\mathbf{X}_N = [\mathbf{x}_2 \quad \mathbf{x}_3 \quad \cdots \quad \mathbf{x}_N] \in R^{M \times N-1}.$$

$$\mathbf{X}_N = \mathbf{A}_d \mathbf{X}_{N-1}$$

DMD:

Truncated SVD:

$$\mathbf{X}_{N-1} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$$

Approximated Evolution Matrix with SVD:

$$\tilde{\mathbf{A}} \triangleq \mathbf{U}^* \mathbf{X}_N \mathbf{V} \mathbf{\Sigma}^{-1}$$

By EVD: $\tilde{\mathbf{A}} \phi = \lambda_d \phi$

Estimated Discrete-Time Eigenvalue and DMD mode related to λ_d :

$$\kappa = \ln(\lambda_d) / \Delta t$$

$$\phi \triangleq \frac{1}{\lambda} \mathbf{Y} \mathbf{V} \mathbf{\Sigma}^{-1} \phi$$

Consider a two state Snapshot Matrix:

$$\mathbf{X} = [\Delta \omega \quad \Delta P_e]^T \in R^{2 \times N}.$$

DMD -

Estimated State Evolution Matrix with EVD:

$$\mathbf{A}_c = \ln(\tilde{\mathbf{A}}_d) / \Delta t.$$

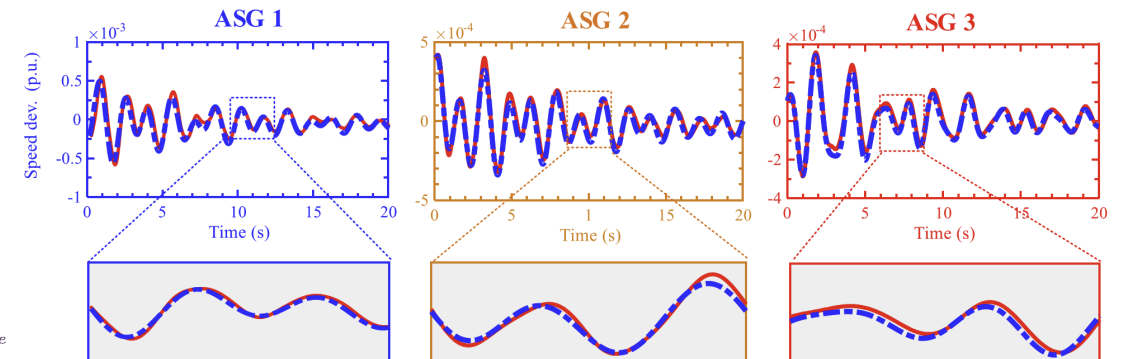
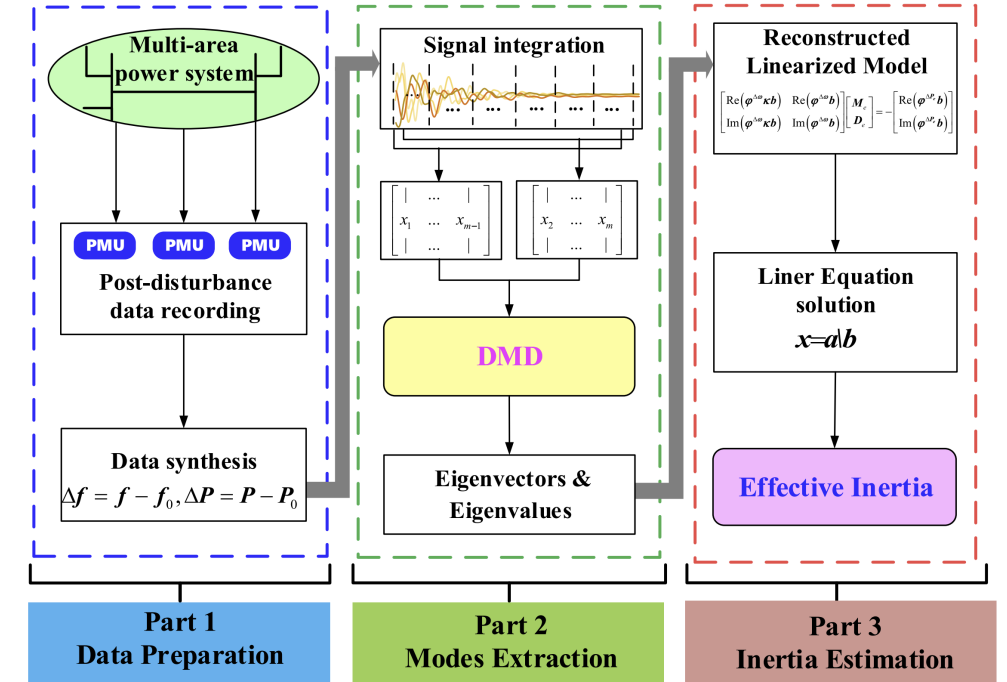
$$\mathbf{A}_c \begin{bmatrix} \varphi_i^{\Delta \omega} \\ \varphi_i^{\Delta P_e} \end{bmatrix} = \kappa_i \begin{bmatrix} \varphi_i^{\Delta \omega} \\ \varphi_i^{\Delta P_e} \end{bmatrix}$$

Reconstructed Snapshot:

$$\begin{bmatrix} \Delta \omega^R \\ \Delta P_e^R \end{bmatrix} = \sum_{i=1}^m e^{\lambda_i t} b_i \begin{bmatrix} \varphi_i^{\Delta \omega} \\ \varphi_i^{\Delta P_e} \end{bmatrix}$$

$$\boxed{M} \sum_{i=1}^m \kappa_i e^{\kappa_i t} b_i \varphi_i^{\Delta \omega} + D \sum_{i=1}^m e^{\kappa_i t} b_i \varphi_i^{\Delta \omega} = - \sum_{i=1}^m e^{\kappa_i t} b_i \varphi_i^{\Delta P_e}$$

Deyou Yang (2021): Data Driven Approach to Extract System Mode



Inertia Estimation – System Identification Approach

Peter Makolo (2022):

Physical Model (Swing Equation):

$$G(s) = \frac{\Delta f}{\Delta P_e} = - \frac{1}{2Hs + D}$$

$$\Delta f(t) = \left(1 - e^{-\frac{D}{M}t} \right) \frac{\Delta P}{D} \quad M = 2H$$

$$H = \frac{\Delta P}{2 \frac{\Delta f}{\Delta t}} \quad \text{at } \frac{d\Delta f}{dt} \Big|_{t=0}$$

System of Linear Equation:

$$\begin{bmatrix} y(n) \\ \vdots \\ y(l-1) \end{bmatrix} = \begin{bmatrix} -y(n-1) & \dots & -y(0) & u(n-1) & \dots & u(0) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ -y(l-2) & \dots & -y(l-n-1) & u(l-2) & \dots & u(l-n-1) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \\ b_{n-m} \\ \vdots \\ b_n \end{bmatrix}$$

Fat Matrix

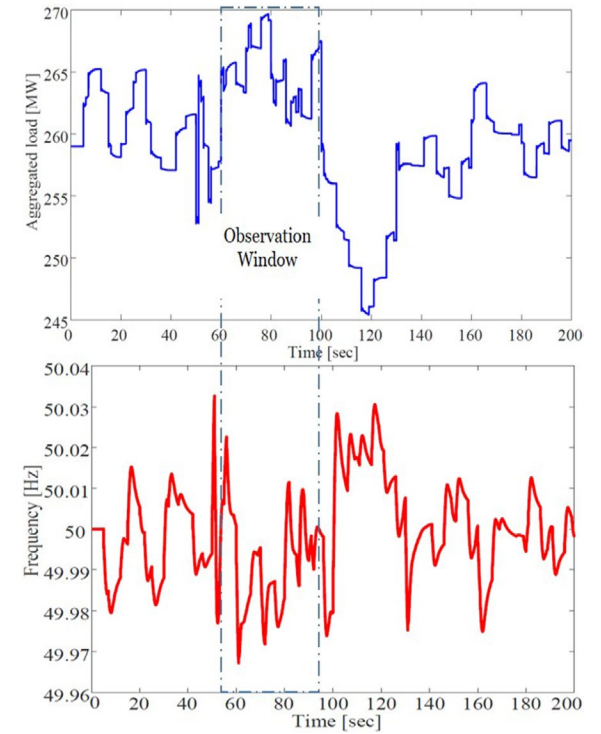
$$Y_N = X_N \Theta$$

Normal Form / Least Square Solution -

$$\Theta^* = X_N^+ Y_N = (X_N^T X_N)^{-1} X_N^T Y_N = U \Sigma V^T = [U_1 \ U_2] \cdot \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \cdot \begin{bmatrix} V_1^T \\ V_2^T \end{bmatrix}$$

Truncated System -

$$G'(s) = \frac{b_{n-1}s^{\sigma_1-1} + b_{n-2}s^{\sigma_1-2} + \dots + b_0}{a_n s^{\sigma_1} + a_{n-1}s^{\sigma_1-1} + \dots + a_0}$$



Exp	Actual value, H (s)	Estimated inertia constant values, $\bar{H}$ (s)		
		Mean	St. Dev.	% Error
1	3.87	4.21	0.51	8.79
2	4.65	4.32	0.54	7.10
3	5.56	5.33	0.59	4.14
4	7.69	7.19	0.50	6.50
5	8.12	8.66	0.53	6.65
6	9.21	9.79	0.47	6.30
7	10.02	10.59	0.31	5.69

Estimated Model (Discrete): **E.O. Kontis (2021): ARMAX Model**

$$G(z) = \frac{b_1 z + b_2 z^2 + \dots + b_m z^m}{1 + a_1 z + \dots + a_{n-1} z^{n-1} + a_n z^n}$$

$$y(k) + a_1 y(k-1) + \dots + a_n y(k-n) = b_1 u(k-1) + \dots + b_m u(k-m)$$

Step Response to
Truncated System -

Sliding Window Approach -

$$Gradient_1 = \frac{1}{A} \sum_{t=0}^{t_e} \frac{d\Delta f(t)}{dt} \quad H = \frac{1}{2 \times Gradient_1}$$

Inertia Estimation – System Identification with Recursive Least Square (RLS)

Ma (2022): FF-RLS

Estimated Model (ARMAX):

$$A(z)y(t) = B(z)u(t) + v(t)$$

$$\begin{cases} A(z) = 1 + a_1(t)z^{-1} + \dots + a_{n_a}(t)z^{-n_a} \\ B(z) = b_0(t) + b_1(t)z^{-1} + \dots + b_{n_b}(t)z^{-n_b} \end{cases}$$

$$\theta = [a_1, a_2, \dots, a_{n_a}, b_1, b_2, \dots, b_{n_b}]^T$$

$$\varphi(t) = [-y(t-1), \dots, -y(t-n_a), u(t-1), \dots, u(t-n_b)]^T$$

Model in Least Square Format:

$$y(t) = \varphi^T(t)\theta + v(t)$$

Stacked Model: $t = 1, 2, 3, \dots, N$ (N = Data Length, $N \gg n$)

$$Y_N = H_N \theta + V_N$$

$$Y_N = [y(1), y(2), \dots, y(N)]^T$$

$$H_N = [\varphi^T(1), \varphi^T(2), \dots, \varphi^T(N)]^T$$

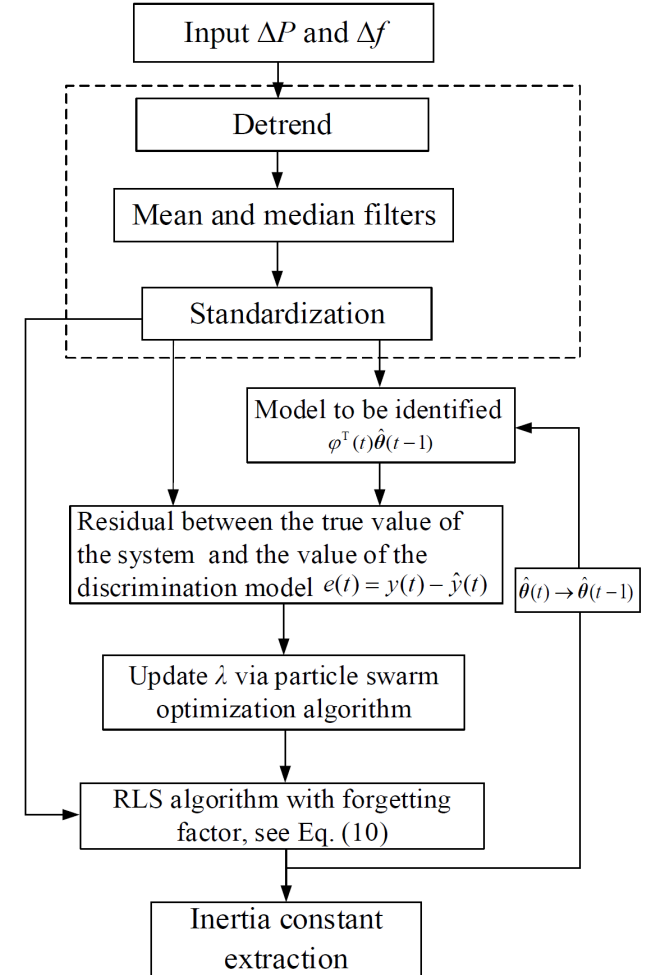
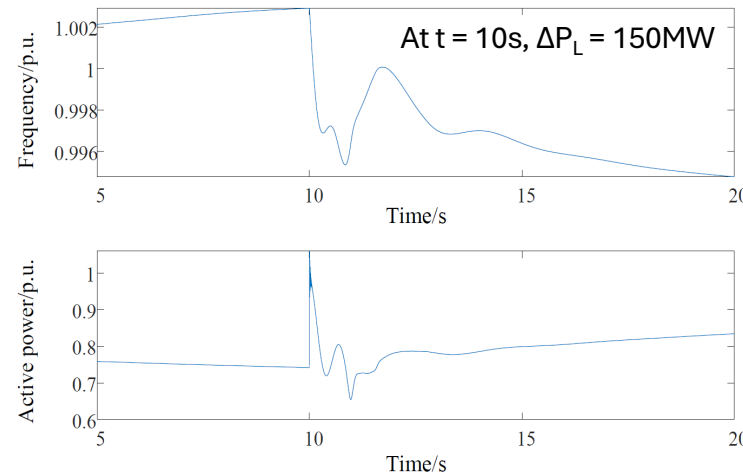
$$V_N = [v(1), v(2), \dots, v(N)]^T$$

Recursive Least Square (RLS) Algorithm:

$$\begin{cases} \hat{\theta}(t) = \hat{\theta}(t-1) + L(t)[y(t) - \varphi^T(t)\hat{\theta}(t-1)] \\ L(t) = P(t-1)\varphi(t)[1 + \varphi^T(t)P(t-1)\varphi(t)]^{-1} \\ P(t) = [I - L(t)\varphi^T(t)]P(t-1) \end{cases}$$

RLS with Forgetting Factor:

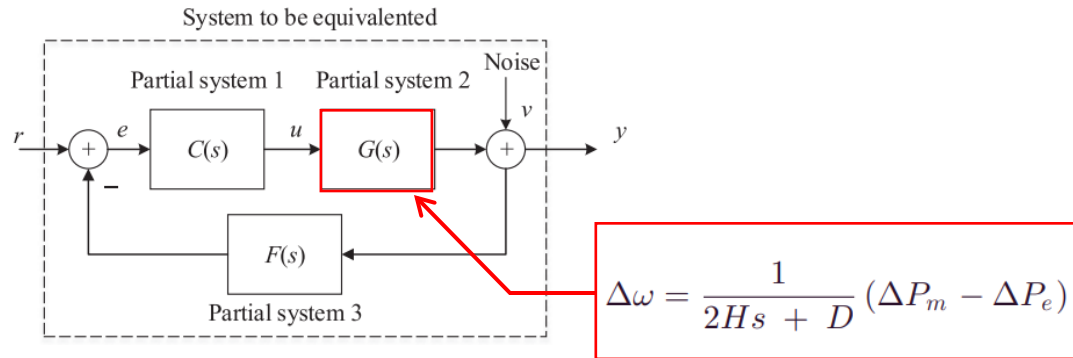
$$\begin{cases} \hat{\theta}(t) = \hat{\theta}(t-1) + L(t)[y(t) - \varphi^T(t)\hat{\theta}(t-1)] \\ L(t) = P(t-1)\varphi(t)[\lambda + \varphi^T(t)P(t-1)\varphi(t)]^{-1} \\ P(t) = \frac{1}{\lambda}[I - L(t)\varphi^T(t)]P(t-1) \end{cases}$$



Micro-perturbation Method – Active Injection

Zhang (2017): MPM

Consider a micro-perturbation $r(t)$ for a closed-loop identification -



$$y(s) = G(s)u(s) + v(s)$$

$$u(s) = C(s)e(s)$$

$$e(s) = r(s) - F(s)y(s)$$

$$u(s) = \frac{C(s)}{1 + C(s)G(s)F(s)}r(s) - \frac{C(s)F(s)}{1 + C(s)G(s)F(s)}v(s)$$

$$y(s) = \frac{C(s)G(s)}{1 + C(s)G(s)F(s)}r(s) + \frac{1}{1 + C(s)G(s)F(s)}v(s)$$

$$\frac{y(s)}{u(s)} = \frac{G(s) + \frac{1}{C(s)} \frac{v(s)}{r(s)}}{1 - F(s) \frac{v(s)}{r(s)}}$$

$G(s)$ is identifiable when the energy of $r(t)$ is larger than that of $v(t)$,
i.e. higher SNR for accurate estimation.
cannot be identified in closed loop

Micro-perturbation Method (MPM) suggests a multi-sine signal:

$$r(k) = \sum_{j=1}^{N_j} a_j \sin(w_j k + \varphi_j)$$

with user assigned ω_j , find a set of $\{a_j, \omega_j, \phi_j\}$ which maximize a_j with bounded $\max(|r(k)|)$. It is solvable with p-Chebyshev approximation method.

Orthogonal Decomposition based Subspace Method (ORT):

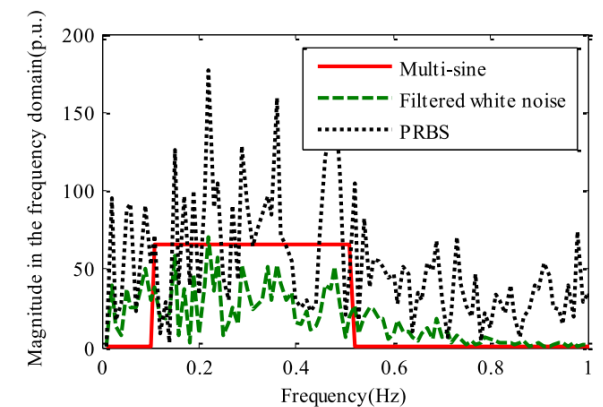
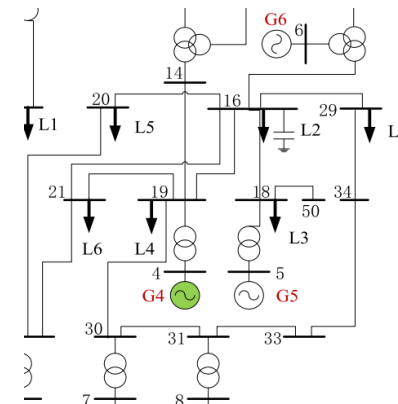
Minimal Realization
Algorithm

$$T_{ru}(s) = \frac{u(s)}{r(s)} \approx \frac{C(s)}{1 + C(s)G(s)F(s)}$$

$$T_{ry}(s) = \frac{y(s)}{r(s)} \approx \frac{C(s)G(s)}{1 + C(s)G(s)F(s)}$$

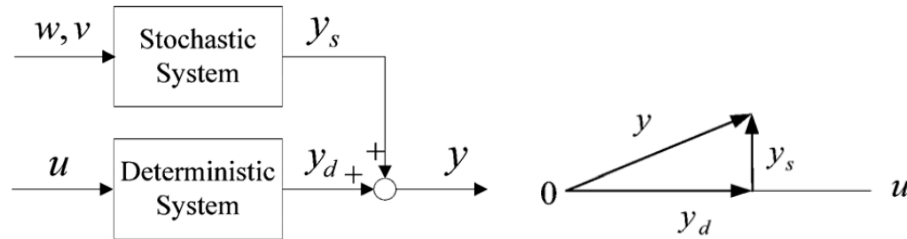
$$\hat{G}(s) = \frac{T_{ry}(s)}{T_{ru}(s)}$$

$$G(s) = \frac{\Delta\omega}{\Delta P_{eo}} \approx -\frac{1}{2Hs + D}$$



Micro-perturbation Method – Active Injection

Orthogonal Decomposition based Subspace Method (ORT): **Zhang (2013)**



MIMO State Space Model:

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k) + \mathbf{w}(k) \\ \mathbf{y}(k) &= \mathbf{C}\mathbf{x}(k) + \mathbf{D}\mathbf{u}(k) + \mathbf{v}(k) \end{aligned} \longrightarrow \mathbf{y}(k) = \mathbf{y}_d(k) + \mathbf{y}_s(k)$$

$$E \left\{ \begin{bmatrix} \mathbf{w}(k) \\ \mathbf{v}(k) \end{bmatrix} [\mathbf{w}^T(s) \quad \mathbf{v}^T(s)] \right\} = \begin{bmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}^T & \mathbf{R} \end{bmatrix} \delta_{ks}$$

$$\mathbf{y}_d(k) \perp \mathbf{y}_s(\tau), \quad k, \tau = 0, \pm 1, \dots$$

$\mathbf{y}_d$ = deterministic signal
 $\mathbf{y}_s$ = stochastic signal

Consider an n-th order deterministic systems:

$$\begin{aligned} \mathbf{x}_d(k+1) &= \mathbf{A}_d \mathbf{x}_d(k) + \mathbf{B}_d \mathbf{u}(k) \\ \mathbf{y}_d(k) &= \mathbf{C}_d \mathbf{x}_d(k) + \mathbf{D}_d \mathbf{u}(k) \end{aligned}$$

1. Collect a sequence of output and input vector to form the Hankel Matrix

$$\begin{aligned} \{\mathbf{y}(0), \mathbf{y}(1), \dots, \mathbf{y}(N+k-2)\} \\ \{\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(N+k-2)\} \end{aligned} \quad \mathbf{U}_{0|k-1} = \begin{bmatrix} u(0) & u(1) & \cdots & u(N-1) \\ \vdots & \vdots & \ddots & \vdots \\ u(k-1) & u(k) & \cdots & u(N+k-2) \end{bmatrix}$$

$$\mathbf{U}_{k|2k-1} = \begin{bmatrix} u(k) & u(k+1) & \cdots & u(k+N-1) \\ \vdots & \vdots & \ddots & \vdots \\ u(2k-1) & u(2k) & \cdots & u(N+2k-2) \end{bmatrix}$$

2. Perform QR Factorization to the joint in-out Hankel matrix

$$\begin{bmatrix} \mathbf{U}_{k|2k-1} \\ \mathbf{U}_{0|k-1} \\ \mathbf{Y}_{0|k-1} \\ \mathbf{Y}_{k|2k-1} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_{11} & & & \\ \mathbf{L}_{21} & \mathbf{L}_{22} & & \\ \mathbf{L}_{31} & \mathbf{L}_{32} & \mathbf{L}_{33} & \\ \mathbf{L}_{41} & \mathbf{L}_{42} & \mathbf{L}_{43} & \mathbf{L}_{44} \end{bmatrix} \begin{bmatrix} \mathbf{Q}_1^T \\ \mathbf{Q}_2^T \\ \mathbf{Q}_3^T \\ \mathbf{Q}_4^T \end{bmatrix}$$

Deterministic systems to be extracted with $\mathbf{L}_{42}$

3. Perform SVD on the submatrix $\mathbf{L}_{42}$ and determine the extended observability matrix.

$$\mathbf{L}_{42} = [\hat{\mathbf{U}} \quad \bar{\mathbf{U}}] \begin{bmatrix} \hat{\mathbf{S}} & \mathbf{0} \\ \mathbf{0} & \bar{\mathbf{S}} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{V}}^T \\ \bar{\mathbf{V}}^T \end{bmatrix} \approx \hat{\mathbf{U}} \hat{\mathbf{S}} \hat{\mathbf{V}}^T \longrightarrow \mathbf{O}_k = \hat{\mathbf{U}} \hat{\mathbf{S}}^{1/2}$$

$$\mathbf{O}_k \triangleq \begin{bmatrix} \mathbf{C}_d \\ \mathbf{C}_d \mathbf{A}_d \\ \vdots \\ \mathbf{C}_d \mathbf{A}_d^{k-1} \end{bmatrix}$$

4. Estimate $\mathbf{A}_d$ and $\mathbf{C}_d$ by -

$$\mathbf{A}_d = \mathbf{O}_{k-1}^+ \bar{\mathbf{O}}_k \quad \mathbf{C}_d = \mathbf{O}_k(1:p, :)$$

5. Consider $\bar{\mathbf{U}}$ as the orthogonal complement of $\mathbf{O}_k$, then $\mathbf{B}_d$ and $\mathbf{D}_d$ can be calculated from the equations with least square.

$$\bar{\mathbf{U}}^T \mathbf{L}_{41} \mathbf{L}_{11}^{-1} = \bar{\mathbf{U}}^T \boldsymbol{\Psi}_k(\mathbf{B}_d, \mathbf{D}_d) \quad \boldsymbol{\Psi}_k \triangleq \begin{bmatrix} \mathbf{D}_d & & & \\ \mathbf{C}_d \mathbf{B}_d & \mathbf{D}_d & & \\ \mathbf{C}_d \mathbf{A}_d \mathbf{B}_d & \mathbf{C}_d \mathbf{B}_d & \mathbf{D}_d & \\ \vdots & \vdots & \ddots & \ddots \\ \mathbf{C}_d \mathbf{A}_d^{k-2} \mathbf{B}_d & \cdots & \cdots & \mathbf{C}_d \mathbf{B}_d & \mathbf{D}_d \end{bmatrix}$$

$$\mathbf{Y}_{0|k-1} = \begin{bmatrix} y(0) & y(1) & \cdots & y(N-1) \\ \vdots & \vdots & \ddots & \vdots \\ y(k-1) & y(k) & \cdots & y(N+k-2) \end{bmatrix}$$

$$\mathbf{Y}_{k|2k-1} = \begin{bmatrix} y(k) & y(k+1) & \cdots & y(k+N-1) \\ \vdots & \vdots & \ddots & \vdots \\ y(2k-1) & y(2k) & \cdots & y(N+2k-2) \end{bmatrix}$$

Comparison of Inertia Estimation Algorithm

Category	Algorithm/Key References	Model/Key Features	Advantages	Disadvantages	Remarks
Offline	Swing equation	Uses post-event data (ringdown) and known power imbalance to estimate inertia via RoCoF.	Simple implementation.	Requires exact disturbance size ; sensitive to noise and RoCoF measurement errors.	Limited to short-term frequency changes ; relies on rare transient events.
	Curve fitting	Fits polynomial to frequency transients to filter noise and estimate RoCoF.	Reduces noise impact; robust to oscillations.	Assumes constant power imbalance; depends on event suitability.	Suitable for traditional grids with high synchronous generation.
	RV approach	Accounts for voltage/frequency-dependent load dynamics post-disturbance.	More realistic by including load dynamics.	Complex; requires accurate load models.	Improves accuracy but still event-dependent.
Online	Sliding Window Method (SWM)	Uses sliding data windows to estimate inertia and disturbance time simultaneously.	Fast computation; suitable for real-time.	Optimal window size critical; sensitive to noise.	Works for both ambient and ringdown data.
	ARMAX	System identification using autoregressive models with exogenous inputs.	Handles ambient data; captures unknown power changes.	Requires model order reduction; computational complexity.	Robust under RES penetration but affected by fault location.
	Electromechanical Oscillations	Estimates inertia from modal parameters (frequency, damping) of oscillations.	No need for disturbance; works with ambient data.	Fails with highly damped or nonstationary oscillations.	Suitable for multi-area systems but sensitive to noise.
	Micro-Perturbation Method (MPM)	Injects probing signals (e.g., multisine) to estimate inertia.	High signal-to-noise ratio; accurate for online use.	Requires additional hardware; complex implementation.	Better than transient-based methods but limited to local buses.
	Kalman Filter	Estimates rotor speed and inertia using PMU data and dynamic models.	Handles noise; tracks time-varying inertia.	Requires accurate state-space modeling; convergence issues.	Decentralized; suitable for individual generator estimation.
	Regression-based	Combines generator status and load forecasts for inertia prediction.	Includes load-side contributions.	Underestimates without embedded generation data.	Adopted by ERCOT and National Grid.

Extended Kalman Filter (EKF) vs Iterative Extended Kalman Filter (IEKF)

Extended Kalman Filter (EKF)

Time Update – “Predict”

- Extrapolate the state

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1}, u_{k-1})$$

- Extrapolate the uncertainty

$$P_{k|k-1} = F_k P_{k-1} F_k^T + Q_k, \quad F_k = \left. \frac{\partial f}{\partial x} \right|_{x_{k-1}}$$

Measurement Update – “Correct”

- Innovation:

$$y_k = z_k - h(\hat{x}_{k|k-1})$$

- Kalman gain:

$$K_k = P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1}, \quad H_k = \left. \frac{\partial h}{\partial x} \right|_{\hat{x}_{k|k-1}}$$

- State update:

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k y_k$$

- Covariance update:

$$P_k = (I - K_k H_k) P_{k|k-1}$$

Iterative Extended Kalman Filter (IEKF)

Measurement Update – “Correct”

- Initialize:

$$\hat{x}_k^{(0)} = \hat{x}_{k|k-1}$$

Repeat until convergence $\|\hat{x}_k^{(i+1)} - \hat{x}_k^{(i)}\| < \varepsilon$:

- Innovation:

$$y_k^{(i)} = z_k - h(\hat{z}_k^{(i)})$$

- Kalman gain:

$$K_k^{(i)} = P_{k|k-1} (H_k^{(i)})^T \left[H_k^{(i)} P_{k|k-1} (H_k^{(i)})^T + R_k \right]^{-1}, \quad H_k^{(i)} = \left. \frac{\partial h}{\partial x} \right|_{x_x^{(0)}}$$

- State update:

$$\hat{x}_k^{(i+1)} = \hat{x}_{k|k-1} + K_k^{(i)} y_k^{(i)}$$

Final state and covariance:

- $\hat{x}_k = \hat{x}_k^{(i+1)}$
- $P_k = (I - K_k^{(i)} H_k^{(i)}) P_{k|k-1}$

Extended Kalman Filter (EKF) – Theory

$$\begin{cases} \frac{d\mathbf{x}}{dt} = f_c(\mathbf{x}, \mathbf{y}, \mathbf{u}, \mathbf{w}) \\ 0 = g_c(\mathbf{x}, \mathbf{y}, \mathbf{u}, \mathbf{v}) \end{cases}$$

- Dynamic Equation (Nonlinear)

- Algebraic Equation (Nonlinear)

$\mathbf{x}$ = state variables (dynamics)

$\mathbf{y}$ = algebraic variables

$\mathbf{u}$ = input

$\mathbf{w}$ = processing noise (related to $\mathbf{x}$)

$\mathbf{v}$ = measurement noise (related to $\mathbf{y}$)

$$\begin{cases} \frac{d\delta}{dt} = \omega - \omega_0 \\ \frac{d\omega}{dt} = \frac{\omega_0}{2H} (P_m - P_e - D(\omega - \omega_0)) \\ \frac{d\omega}{dt} = \frac{\omega_0}{2H} \left(P_m - \frac{EV}{X'_d} \sin(\delta - \theta) - D(\omega - \omega_0) \right) \end{cases}$$

$$\begin{cases} P_e = \frac{EV}{X'_d} \sin(\delta - \theta) \\ Q_e = \frac{-V^2 + EV \cos(\delta - \theta)}{X'_d} \end{cases}$$

Discretization:

$$\mathbf{x}_k = \mathbf{x}_{k-1} + f_c(\mathbf{x}_{k-1}, \mathbf{u}_{k-1}, \mathbf{w}_{k-1}) \Delta t \equiv f(\mathbf{x}_{k-1}, \mathbf{u}_{k-1}, \mathbf{w}_{k-1})$$

$$0 = g_c(\mathbf{x}_k, \mathbf{y}_k, \mathbf{u}_k, \mathbf{v}_k) \Rightarrow \mathbf{y}_k = h(\mathbf{x}_k, \mathbf{u}_k, \mathbf{v}_k)$$

$$A = \frac{\partial f}{\partial \mathbf{x}}, \quad H_z = \frac{\partial h}{\partial \mathbf{x}}, \quad W = \frac{\partial f}{\partial \mathbf{w}}, \quad V = \frac{\partial h}{\partial \mathbf{v}}.$$

Model Based -

$$\omega_{k+1} = \frac{\omega_0}{2H_k} \left(P_{m,k} - \frac{EV_k \sin(\delta_k - \theta_k)}{X'_{d,k}} - D_k \left(\frac{\omega_k}{\omega_0} - 1 \right) \right) \Delta t + \omega_k, \quad \text{Method A}$$

$$\omega_{k+1} = \omega_k + \frac{\omega_0}{2H_k} \left(P_{m,k} - P_{e,k} + D_k \left(\frac{\omega_k}{\omega_0} - 1 \right) \right) \Delta t, \quad \text{Method B}$$

Measurement Based -

Prediction :

$$\begin{cases} \hat{\mathbf{x}}_k^- = f(\hat{\mathbf{x}}_{k-1}, h(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_{k-1}), \mathbf{u}_{k-1}, 0) \\ P_k^- = A_k P_{k-1} A_k^T + W_k Q_{k-1} W_k^T \end{cases}$$

State Prediction based on Dynamic Model

Error Covariance Prediction based on Dynamic Model

Correction :

$$\begin{cases} K_k = P_k^- H_{z,k}^T (H_{z,k} P_k^- H_{z,k}^T + V_k R_k V_k^T)^{-1} \\ \hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + K_k (\mathbf{z}_k - h(\hat{\mathbf{x}}_k^-, \mathbf{u}_k, 0)) \\ P_k = (I - K_k H_{z,k}) P_k^- \end{cases}$$

Minimizing Error Covariance with a Proper K

Correct State Prediction with a Proper K and Measurement

Error Correct Covariance Update with Measurement

Extended Kalman Filter (EKF) – Theory

Linearized Model -

$$\begin{cases} \delta_{k+1} = \delta_k + (\omega_k - \omega_0)\Delta t + w_1 \\ \omega_{k+1} = \omega_k + \frac{\omega_0}{2H_k}(P_{m,k} - P_{e,k})\Delta t + D_k(\omega_k - \omega_0)\Delta t + w_2 \\ P_{m,k+1} = P_{m,k} + w_3 \\ H_{k+1} = H_k + w_4 \\ D_{k+1} = D_k + w_5 \\ x'_{d,k+1} = x'_{d,k} + w_6 \end{cases}$$

State Evolution Matrix -

$$A = \begin{bmatrix} 1 & \Delta t & 0 & 0 & 0 & 0 \\ 0 & 1 - \frac{D\omega_0\Delta t}{2H} & \frac{\omega_0\Delta t}{2H} & A_{24} & A_{25} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{24} = -\frac{P_m - P_e - D(\omega - \omega_0)}{2H^2}\omega_0\Delta t$$

$$A_{25} = \frac{-(\omega - \omega_0)}{2H}\omega_0\Delta t$$

Measurement Sensitivity Matrix (for Nonlinear Systems) -

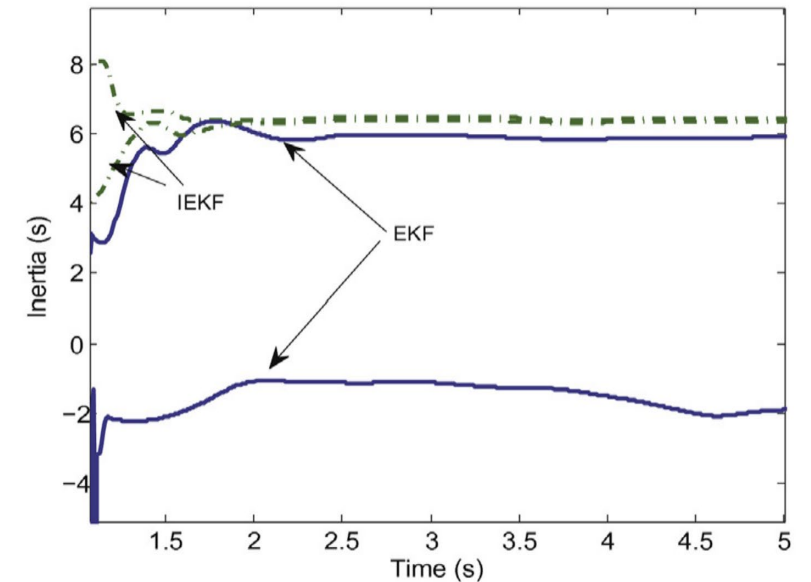
$$H_z = \begin{bmatrix} \frac{\partial V}{\partial \delta} & \frac{\partial V}{\partial \omega} & \frac{\partial V}{\partial P_m} & \frac{\partial V}{\partial H} & \frac{\partial V}{\partial D} & \frac{\partial V}{\partial x'_d} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\partial V}{\partial x'_d} \\ 1 & 0 & 0 & 0 & \frac{\partial \theta}{\partial x'_d} \end{bmatrix}$$

$$\frac{\partial V}{\partial x'_d} = \frac{((-Q_e E^2 - 2x'_d P_e^2)/\sqrt{f_1}) - 4Q_e}{\sqrt{f_2}}$$

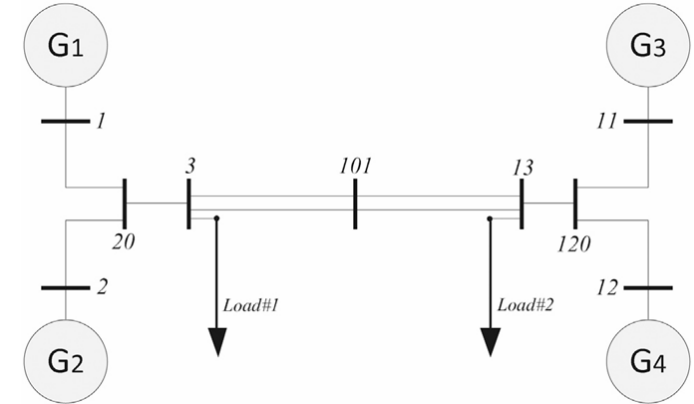
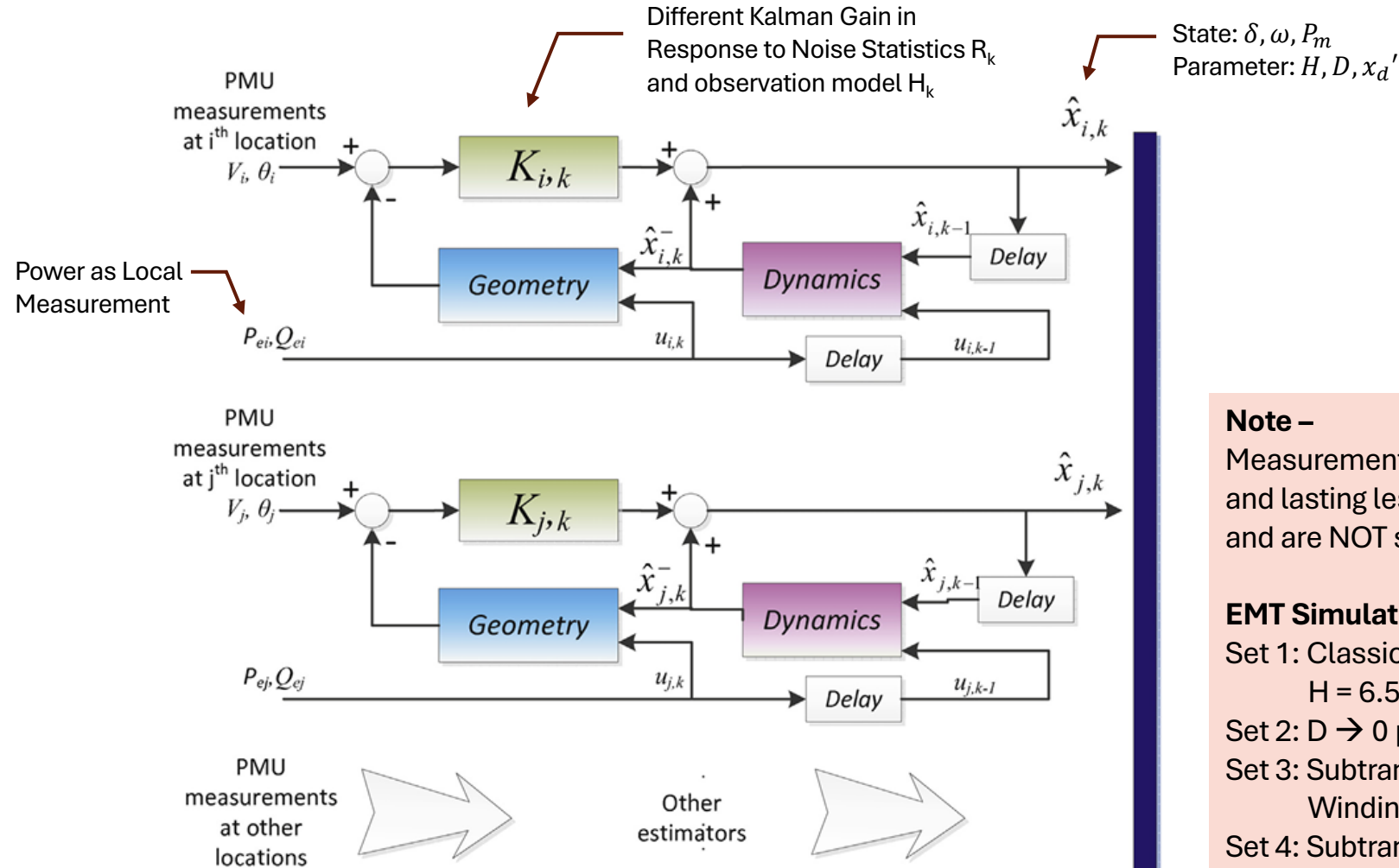
$$\frac{\partial \theta}{\partial x'_d} = \frac{2P_e f_2 - P_e x'_d (((-4Q_e E^2 - 8x'_d P_e^2)/\sqrt{f_1}) - 4Q_e)}{f_2 \sqrt{E^2 f_2 - 4P_e^2 x'^2_d}}$$

**IEKF =
Iteratively Linearize
at Each Steps**

**# Converge with
Acceptable Small
Noise.**



EKF and IEKF in PMU with Simulation -



Four Buses System –

A three phase fault occurred at Bus 3 at $t = 1.0s$,
 Load#1 is tripped $t = 1.1s$.

Note –

Measurements from fault recorders with high sampling rates and lasting less than 1 sec are dominated by EMT dynamics and are NOT suitable for the proposed method

EMT Simulation for Comparison -

Set 1: Classical Generator Model:

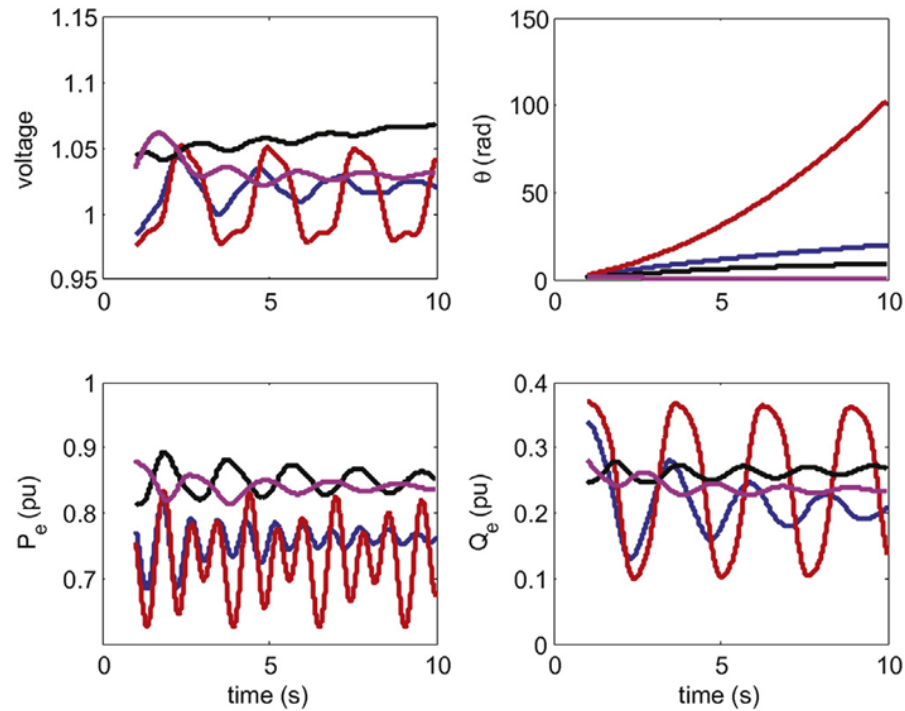
$H = 6.5s$, $D = 6pu$, $x_d' = 0.25pu$, $E = 1.08pu$, $P_m = 0.85pu$

Set 2: $D \rightarrow 0 pu$ in Swing Equation

Set 3: Subtransient Model with Damper Winding and Field Winding (Simulation Model more complex)

Set 4: Subtransient Model with AVR to get stable system response.

Simulation Result -



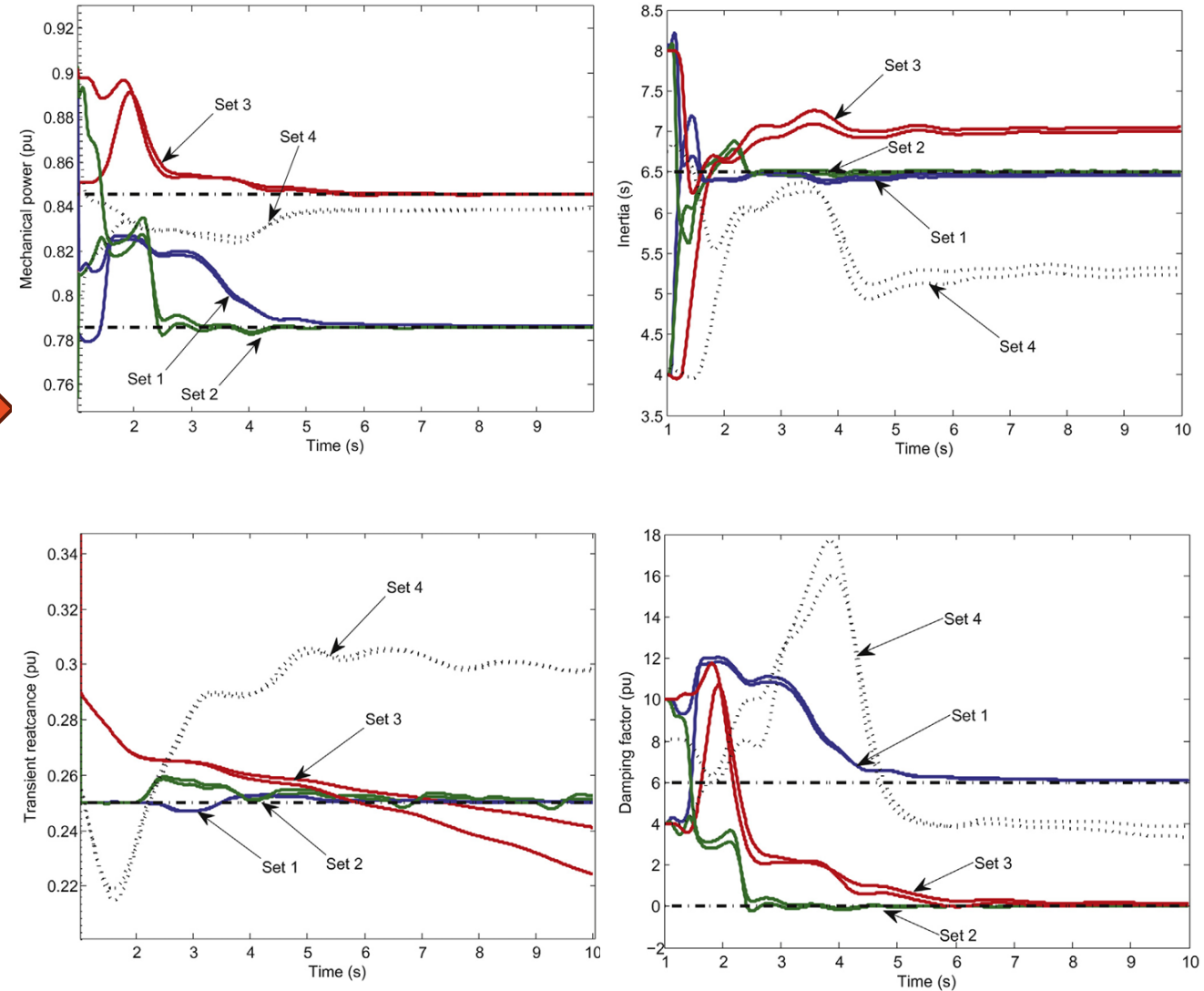
EMT Simulation for Comparison -

Set 1: Classical Generator Model with H, D, x_d', E_0, P_{m0}

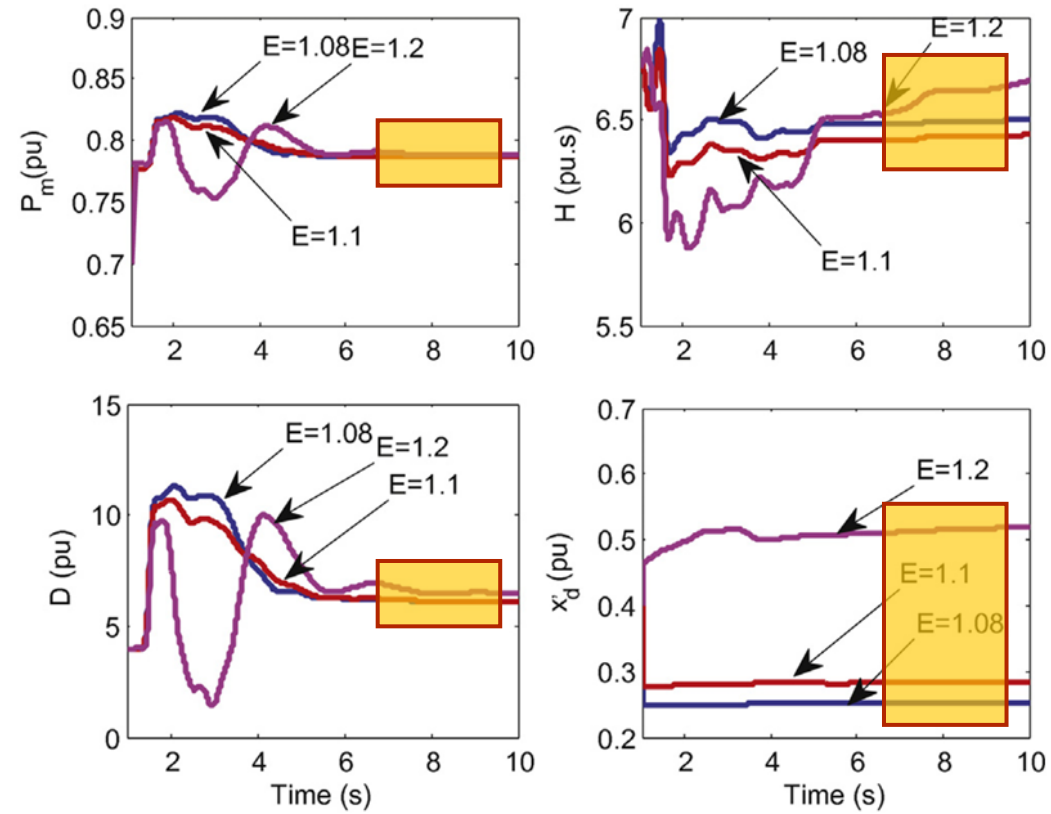
Set 2: Classical Generator Model with $D = 0$

Set 3: Subtransient Model with Damper Winding

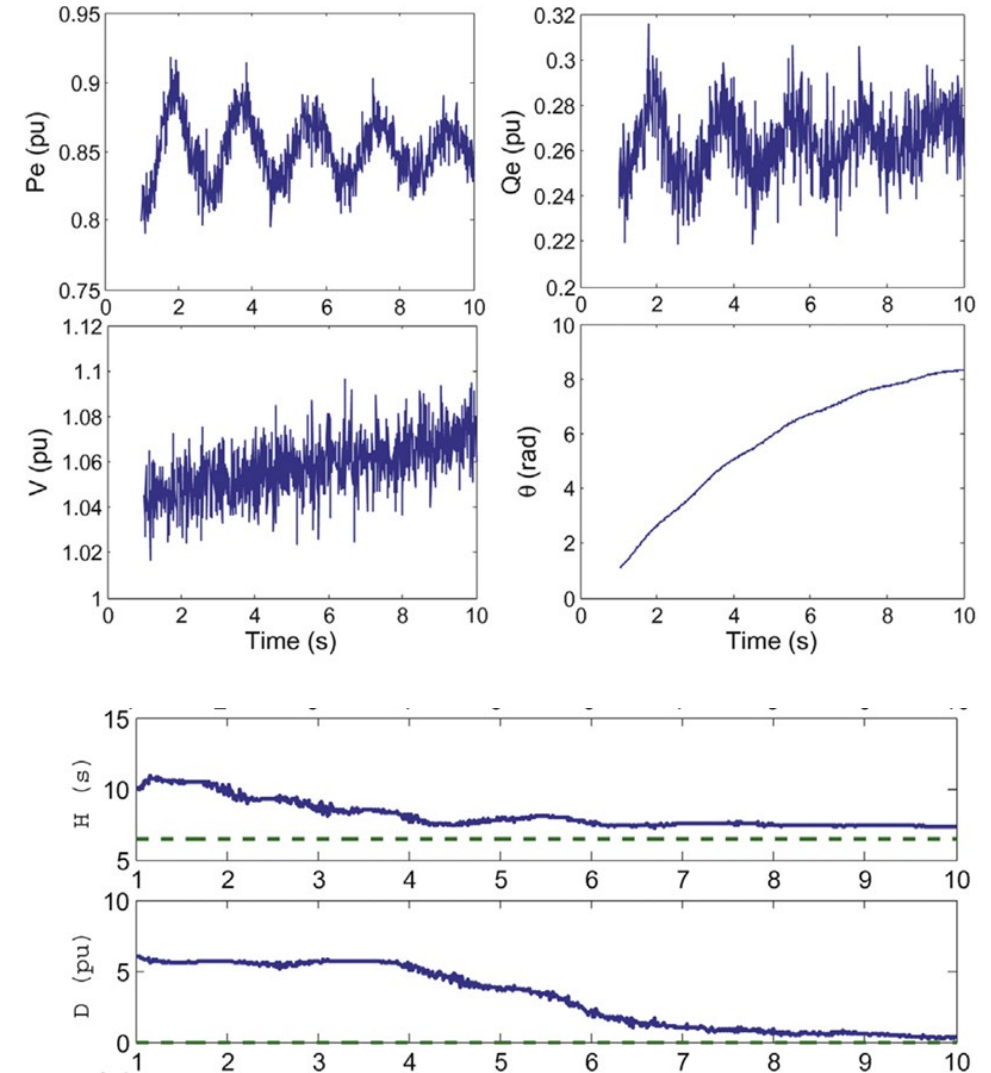
Set 4: Subtransient Model with AVR



Simulation Result -



Effect of Estimation with Different E



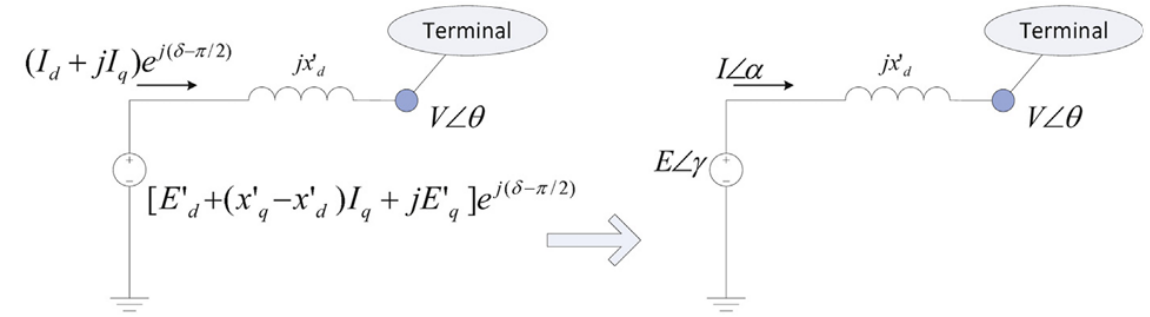
Measurement and Result with White Noise

Filtering and Modelling Highlights -

Assumed Covariance Matrix to cover modelling error and noise -

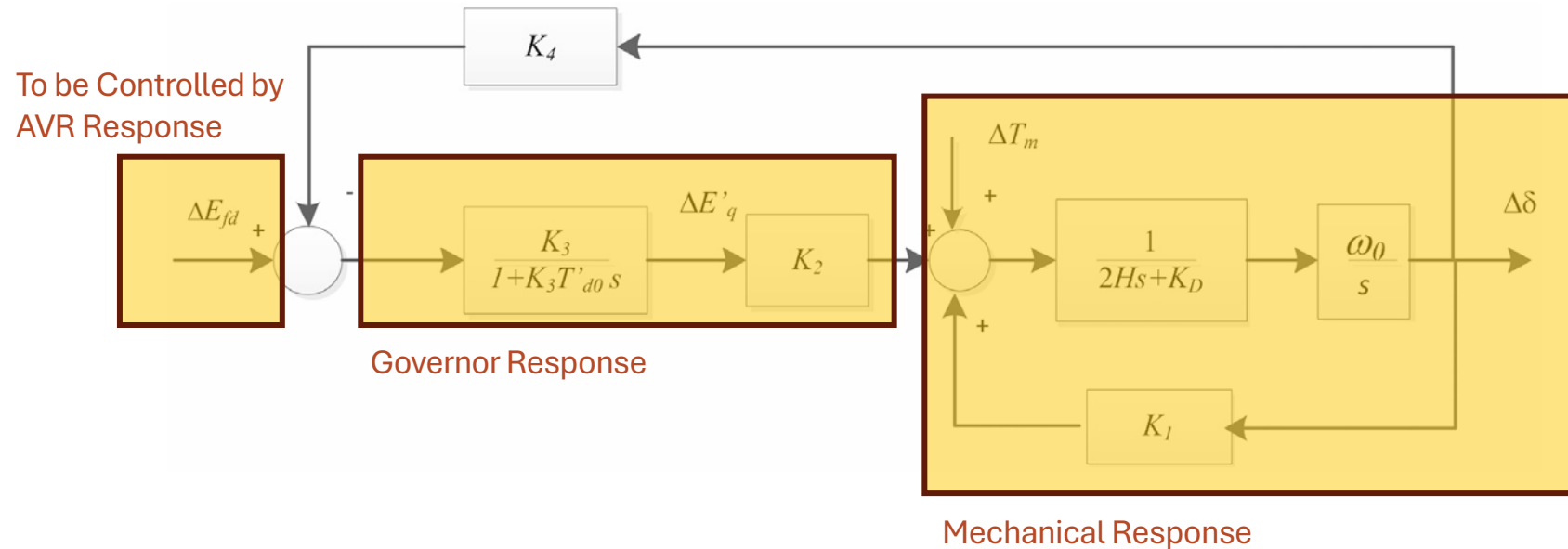
Covariance matrices for two-state four-parameter estimation.

P	Sets 1 and 2	Sets 3 and 4	Q	Sets 1 and 2	Sets 3 and 4
P_{11}	1	1	Q_{11}	$10^{-4} \Delta t$	$10^{-4} \Delta t$
P_{22}	30	30	Q_{22}	$10^{-3} \Delta t$	$10^{-3} \Delta t$
P_{33}	0.1	0.1	Q_{33}	0	0
P_{44}	5	5	Q_{44}	0	0
P_{55}	50	50	Q_{55}	0	0
P_{66}	1	1	Q_{66}	0	$0.01 \Delta t$



Two Axis Model in Control

Traditional Sync Gen Model



Linearized Sync Gen Model
Considering Field Flux Variation

Analysis -

- Linearized Swing Model –

$$s^2(\Delta\delta) + \frac{D}{2H}s(\Delta\delta) + \frac{K_s}{2H}\omega_0(\Delta\delta) = \frac{\omega_0}{2H}\Delta P_m$$

D = Damping Torque
 $K_s = \partial P_e / \partial \delta$ Synchronizing Torque

- The effect of field flux variations will change the synchronizing torque component and the damping torque component by decreasing K_s and increasing K_D .
- Contribution of K_s and K_D due to field flux variation or ΔE_q

$$\frac{\Delta T_e}{\Delta \delta} \bigg|_{\text{due to } \Delta E'_q} = \frac{-K_2 K_3 K_4}{1 + s K_3 T'_{d0}} \longrightarrow \text{Re} \left[\frac{\Delta T_e}{\Delta \delta} \right] = \frac{-K_2 K_3 K_4}{1 + \omega^2 K_3^2 T'^2_{d0}} \quad \text{Im} \left[\frac{\Delta T_e}{\Delta \delta} \right] = \frac{K_2 K_3^2 K_4 T'_{d0}}{1 + \omega^2 K_3^2 T'^2_{d0}}$$

Decreasing K_s due to Armature Reaction Increasing K_D

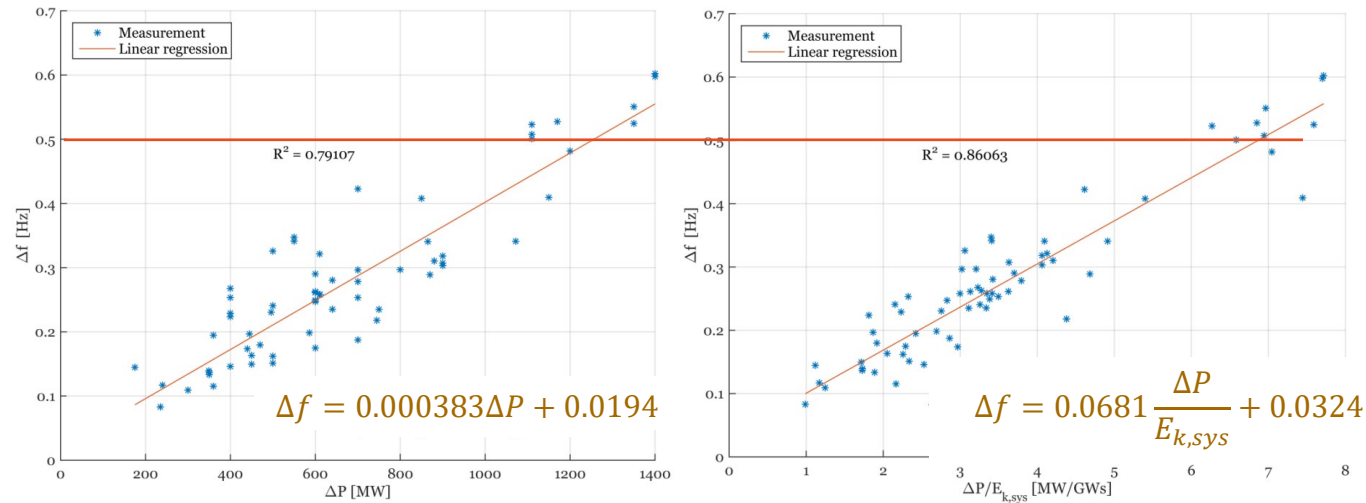
- All parameters except H can be estimated accurately in 10s. The estimation of H is greater than the true value due to modeling error
- Compared to the effect of machine circuit dynamics, the effect of the AVR is much significant.

$$s^2 + \frac{DK_{fD}}{2H}s + \frac{K_s K_{fs} \omega_0}{2H} = 0 \longrightarrow s^2 + \frac{D'}{2H'}s + \frac{K_s \omega_0}{2H'} = 0$$

- In Set 3: $K_{fD} > 1$ (increase in the damping torque component), $K_{fs} < 1$ (reduction in the synchronizing torque component). Hence, $H' = H/K_{fs} > H$, and $D' = DK_{fD}/K_{fs} > D$. → **Overestimate in case AVR response is not considered.**

Inertia Requirement and Estimation in Practice

ENTSOE Check Up Table for Inertia & Contingency Requirement



Difficulties on Inertia Estimation -

1. Reduced Inertia = $\uparrow$ dynamic behaviour = inaccurate estimation
2. Variability of Inertia in the system (as a function of time, $H = H(t)$)
3. Reduction of onset time setting of FFR

System Information

Unknown Generator Parameter

$$E' = - \frac{\Delta P}{2 \frac{d(\Delta f/f_0)}{dt} \big|_{t=0}} \longrightarrow$$

Known Generator Parameter -

$$E = \sum_i H_i P_{i,rated}^G$$

Estimation Frequency

1. **Lower bound** (on-bar generator in SCADA)
2. **Discrete** (PMU to event base under disturbance)
[Swing Equation with Governor Model, Dynamic SG Model, Grid Model]
[Assumption – Constant P_M , LPF leading inaccuracy is insignificant]
3. **Continuous** (PMU even to normal condition)
[Assumption – Power Imbalance > Noise, Generator parameter = constant]

Further Considerations -

1. Accurate and Reliable Inertia Estimate and Forecast in **time-basis** and **regional-basis** contributed from generation, **demand** and **virtual inertia** resources.
2. Contributors at **different operating state**, especially from IBRs uncontrollable and unobservable for TSO.
3. Coordinated and well-localized frequency response **allocations with long links** (weak coupling)
4. **Direct Forecast of frequency deviation and ROCOF** to estimate potential issues in frequency response (e.g. lack of reserves leading to cascade failure)



THE END

ANY QUESTIONS?

Power System Analysis and Protection Studies

Karl M.H. LAI
Department of EEE, HKU & CLP Power Hong Kong
LinkedIn: <https://www.linkedin.com/in/karlmhlai>
Email: karl.lai@clp.com.hk